34

Electromagnetic Waves

CHAPTER OUTLINE

34.1 Displacement Current and the General Form of Ampère's Law

34.2 Maxwell's Equations and Hertz's Discoveries

34.3 Plane Electromagnetic Waves

34.4 Energy Carried by Electromagnetic Waves

34.5 Momentum and Radiation Pressure

34.6 Production of Electromagnetic Waves by an Antenna

34.7 The Spectrum of Electromagnetic Waves

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ34.1 **(i)** Answer (c). Both the light intensity and the gravitational force follow inverse-square laws.

(ii) Answer (a). The smaller grain presents less face area and feels a smaller force due to light pressure.

OQ34.2 **(i)** Answer (c). **(ii)** Answer (c). **(iii)** Answer (c). **(iv)** Answer (b). **(v)** Answer (b). The same amount of energy passes through concentric spheres of increasing area as the wave travels outward from its source, so the amplitude and the intensity, which is proportional to the square of the amplitude, decrease.

OQ34.3 Answer (b). Frequency, wavelength, and the speed of light are related:

$$f\lambda = c \quad \rightarrow \quad \lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{2.45 \times 10^9 \text{ Hz}} = 0.122 \text{ m} = 12.2 \text{ cm}$$

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OQ34.4 **(i)** Answer (a). According to $f = 1/2\pi\sqrt{LC}$, to make f half as large, the capacitance should be made four times larger.

(ii) Answer (b). According to $f\lambda = c$, if frequency is halved, wavelength is doubled.

OQ24.5 Answer (e). Accelerating charge, changing electric field, or changing magnetic field can be the source of a radiated electromagnetic wave.

OQ34.6 Answers (c) and (d). The relationship between frequency, wavelength, and the speed of a wave is $f\lambda = v$. In a vacuum, all electromagnetic waves travel at the same speed: $v = c$. Electromagnetic waves, consisting of oscillating electric and magnetic fields, are transverse waves.

OQ34.7 **(i)** through **(v)** have the same answer (c). The same amount of energy passes through equal areas parallel to the yz plane as the wave travels in the $+x$ direction, so the amplitude and the intensity, which is proportional to the square of the amplitude, do not change.

OQ34.8 **(i)** Answer (b). Electric and magnetic fields both carry the same energy, so their amplitudes are proportional to each other.

(ii) Answer (a). The intensity is proportional to the square of the amplitude.

OQ34.9 Answer (d). The peak values of the electric and magnetic field components of an electromagnetic wave are related by $E_{\max}/B_{\max} = c$, where c is the speed of light in vacuum. Thus,

$$E_{\max} = cB_{\max} = \left(3.00\times10^{8}\ \text{m/s}\right)\left(1.50\times10^{-7}\ \text{T}\right) = 45.0\ \text{N/C}$$

OQ34.10 **(i)** The ranking is c > b > d > e > a. Gamma rays have the shortest wavelength.

(ii) The ranking is a > e > d > b > c. According to $f\lambda = c$, as wavelength decreases, frequency increases.

(iii) The ranking is a = b = c = d = e. All electromagnetic waves travel at the speed of light c in vacuum, which is assumed here.

OQ34.11 Answer (d). An electromagnetic wave travels in the direction of the Poynting vector: $\vec{\mathbf{S}} = \vec{\mathbf{E}} \times \vec{\mathbf{B}}/\mu_0$.

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ANSWERS TO CONCEPTUAL QUESTIONS

CQ34.1 The entire room and its contents have a soft glow. Incandescent light bulbs shine brightly in the infrared, but fluorescent lights do not. The top of a computer monitor glows brighter than the screen, which glows faintly. Windowpanes appear dark if they are cool, and a patch of wall where sunlight falls glows brighter than where the sunlight does not fall. Heating resistors or warm air outlets shine, and the air near to them has a faint glow, but cold air outlets are dark, and the nearby air has no glow.

CQ34.2 Electromagnetic waves carry momentum. Recalling what we learned in Chapter 9, the impulse imparted by a particle that bounces elastically off a wall is twice that imparted by an object that sticks to a wall. Similarly, the impulse, and hence the pressure exerted by a wave reflecting from a surface, must be twice that exerted by a wave that is absorbed.

CQ34.3 No. Radio waves travel at a finite speed, the speed of light. Radio waves can travel around the curved surface of the Earth, bouncing between the ground and the ionosphere, which has an altitude that is small when compared to the radius of the Earth. The distance across the lower forty-eight states is approximately 5 000 km, requiring a transit time of $\frac{5\times10^{6}\ \text{m}}{3\times10^{8}\ \text{m/s}} \sim 10^{-2}\ \text{s}$.

CQ34.4

Sound	*Light*
1) Sound is a longitudinal wave.	1) Light is a transverse wave.
2) Sound requires a material medium.	2) Light does not require a material medium.
3) Sound in air moves at hundreds of meters per second.	3) Light in air moves at hundreds of millions of meters per second.
4) The speed of sound through a medium, depending the material of the medium, can be faster or slower than that in air.	4) The speed of light through materials is less than in vacuum.

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5) Sound propagates by a chain reaction of density and pressure disturbances recreating each other.	5) Light propagates by a chain reaction of electric and magnetic fields recreating each other.
6) Audible sound has frequencies over a range of three decades (ten octaves) from 20 Hz to 20 kHz.	6) Visible light has frequencies over a range of less than one octave, from 430 to 700 THz (THz = Terahertz = 10^{12} Hz).
7) Audible sound has wavelengths of ordinary size (1.7 cm to 17 m).	7) Visible light has wavelengths of very small size (400 nm to 750 nm).

CQ34.5 The changing magnetic field of the solenoid induces eddy currents in the conducting core. This is accompanied by I^2R conversion by heating of electrically-transmitted energy into internal energy in the conductor.

CQ34.6 (a) The electric and magnetic fields of the light wave oscillate in time at each point in space, like sports fans in a grandstand when the crowd does "the wave."

(b) The wave transports energy.

CQ34.7 An infrared photograph records the infrared light *reflected*, but also *emitted* by a person's face. When a person blushes or exercises or becomes excited, warmer areas glow brighter in the infrared. A person's nostrils and the openings of the ear canals are bright; brighter still are just the pupils of the eyes.

CQ34.8 No, they do not. Specifically, Gauss's law in magnetism prohibits magnetic monopoles. If magnetic monopoles existed, then the magnetic field lines would not have to be closed loops, but could begin or terminate on a magnetic monopole, as they can in Gauss's law in electrostatics.

CQ34.9 Different stations have transmitting antennas at different locations. For best reception align your rabbit ears perpendicular to the straight-line path from your TV to the transmitting antenna. The transmitted signals are also polarized. The polarization direction of the wave can be changed by reflection from surfaces—including the atmosphere—and through Kerr rotation—a change in polarization axis when passing through an organic substance. In your home, the plane of polarization is determined by your surroundings, so antennas need to be adjusted to align with the polarization of the wave.

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CQ34.10 Consider a typical metal rod antenna for a car radio. Charges in the rod respond to the electric field portion of the carrier wave. Variations in the amplitude of the incoming radio wave cause the electrons in the rod to vibrate with amplitudes emulating those of the carrier wave. Likewise, for frequency modulation, the variations of the frequency of the carrier wave cause constant-amplitude vibrations of the electrons in the rod but at frequencies that imitate those of the carrier.

CQ34.11 The Poynting vector $\vec{\mathbf{S}}$ describes the energy flow associated with an electromagnetic wave. The direction of $\vec{\mathbf{S}}$ is along the direction of propagation and the magnitude of $\vec{\mathbf{S}}$ is the rate at which electromagnetic energy crosses a unit surface area perpendicular to the direction of $\vec{\mathbf{S}}$.

CQ24.12 The frequency of EM waves in a microwave oven, typically 2.45 GHz, is chosen to be in a band of frequencies absorbed by water molecules. The plastic and the glass contain no water molecules. Plastic and glass have very different absorption frequencies from water, so they may not absorb any significant microwave energy and remain cool to the touch.

CQ34.13 Maxwell included a term in Ampère's law to account for the contributions to the magnetic field by changing electric fields, by treating those changing electric fields as "displacement currents."

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 34.1 Displacement Current and the Generalized Form of Ampère's Law

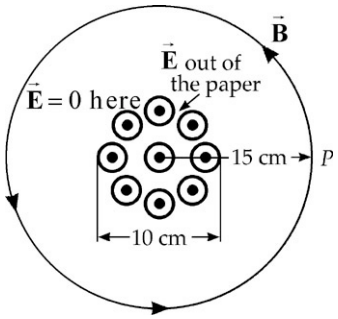
P34.1 (a) We use the right-hand rule for both real and displacement currents. Thus, the direction of $\vec{\mathbf{B}}$ is *counterclockwise*, and the direction of $\vec{\mathbf{B}}$ at P is upwards.



(b) We use the extended form of Ampère's law, Equation 34.7. Since no moving charges are present, $I = 0$ and we have

$$\oint \vec{\mathbf{B}} \cdot d\vec{\ell} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

In order to evaluate the integral, we make use of the symmetry of the



ANS. FIG. P34.1

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situation. Symmetry requires that no particular direction from the center can be any different from any other direction. Therefore, there must be *circular symmetry* about the central axis. We know the magnetic field lines are circles about the axis. Therefore, as we travel around such a magnetic field circle, the magnetic field remains constant in magnitude. Setting aside until later the determination of the *direction* of $\vec{\mathbf{B}}$, we integrate $\oint \vec{\mathbf{B}} \cdot d\vec{\ell}$ around the circle at $R = 0.150$ m to obtain $2\pi RB$.

Differentiating the expression $\Phi_E = AE$, we have

$$\frac{d\Phi_E}{dt} = \left(\frac{\pi d^2}{4}\right)\frac{dE}{dt}$$

Thus, $\oint \vec{\mathbf{B}} \cdot d\vec{\ell} = 2\pi RB = \mu_0 \epsilon_0 \left(\frac{\pi d^2}{4}\right)\frac{dE}{dt} \quad \rightarrow \quad B = \frac{\mu_0 \epsilon_0 d^2}{8R}\frac{dE}{dt}$

Substituting numerical values,

$$B = \frac{\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(8.85\times 10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)(0.100\ \text{m})^2}{8(0.150\ \text{m})} \times (20.0\ \text{V/m}\cdot\text{s})$$

$$= \boxed{1.85\times 10^{-18}\ \text{T}}$$

P34.2 For the capacitor,

$$\frac{d\Phi_E}{dt} = \frac{d}{dt}(EA) = \frac{dQ/dt}{\epsilon_0} = \frac{I}{\epsilon_0}$$

(a) $$\frac{dE}{dt} = \frac{I}{\epsilon_0 A} = \frac{0.200\ \text{A}}{\left(8.85\times 10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)\left[\pi\left(10.0\times 10^{-2}\ \text{m}\right)\right]}$$

$$= \boxed{7.19\times 10^{11}\ \text{V/m}\cdot\text{s}}$$

(b) $$\oint B\cdot ds = \epsilon_0\mu_0\frac{d\Phi_E}{dt}: \qquad 2\pi rB = \cancel{\epsilon_0}\mu_0\frac{d}{dt}\left[\frac{Q}{\cancel{\epsilon_0}A}\cdot\pi r^2\right]$$

$$B = \frac{\mu_0 Ir}{2A} = \frac{\mu_0(0.200\ \text{A})\left(5.00\times 10^{-2}\ \text{m}\right)}{2\left[\pi\left(10.0\times 10^{-2}\ \text{m}\right)^2\right]} = \boxed{2.00\times 10^{-7}\ \text{T}}$$

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P34.3 The electric field in the space between the plates is $E = \dfrac{\sigma}{\epsilon_0} = \dfrac{Q}{\epsilon_0 A}$.

The flux of this field is $\Phi_E = \vec{\mathrm{E}} \cdot \vec{\mathrm{A}} = \left(\dfrac{Q}{\epsilon_0 A}\right) A \cos 0^\circ = \dfrac{Q}{\epsilon_0}$.

(a) The rate of change of flux is

$$\frac{d\Phi_E}{dt} = \frac{dQ/dt}{\epsilon_0} = \frac{I}{\epsilon_0} = \frac{0.100\ \text{A}}{8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2}$$

$$= \boxed{11.3\times10^{9}\ \text{V}\cdot\text{m/s}}$$

(b) The displacement current is defined as

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

$$= (8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2)(1.13\times10^{10}\ \text{N}\cdot\text{m}^2/\text{C}\cdot\text{s})$$

$$= \boxed{0.100\ \text{A}}$$

Section 34.2 Maxwell's Equations and Hertz's Discoveries

P34.4 $\vec{\mathbf{F}} = m\vec{\mathbf{a}} = q\vec{\mathbf{E}} + q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$ so

$$\vec{\mathbf{a}} = \frac{-e}{m}\left[\vec{\mathbf{E}} + \vec{\mathbf{v}} \times \vec{\mathbf{B}}\right] \text{ where}$$

$$\vec{\mathbf{v}} \times \vec{\mathbf{B}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 10.0 & 0 & 0 \\ 0 & 0 & 0.400 \end{vmatrix} = -(4.00\ \text{T}\cdot\text{m/s})\hat{\mathbf{j}}$$

Then

$$\vec{\mathbf{a}} = \left(\frac{-1.60\times10^{-19}\ \text{C}}{9.11\times10^{-31}\ \text{kg}}\right)$$

$$\times\left[(2.50\ \text{V/m})\hat{\mathbf{i}} + (5.00\ \text{V/m})\hat{\mathbf{j}} - (4.00\ \text{T}\cdot\text{m/s})\hat{\mathbf{j}}\right]$$

$$= \left(-1.76\times10^{11}\right)\left[2.50\hat{\mathbf{i}} + 1.00\hat{\mathbf{j}}\right]\ \text{m/s}^2$$

$$\vec{\mathbf{a}} = \boxed{\left(-4.39\hat{\mathbf{i}} - 1.76\hat{\mathbf{j}}\right)\times10^{11}\ \text{m/s}^2}$$

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P34.5 The net force on the proton is the Lorentz force, as described by

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}} = q\vec{\mathbf{E}} + q\vec{\mathbf{v}} \times \vec{\mathbf{B}} \quad \text{so that} \quad \vec{\mathbf{a}} = \frac{e}{m}\left[\vec{\mathbf{E}} + \vec{\mathbf{v}} \times \vec{\mathbf{B}}\right]$$

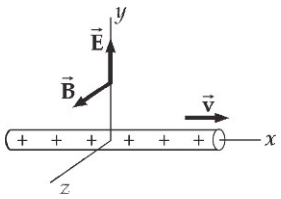
Taking the cross product of $\vec{\mathbf{v}}$ and $\vec{\mathbf{B}}$,

$$\vec{\mathbf{v}} \times \vec{\mathbf{B}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 200 & 0 & 0 \\ 0.200 & 0.300 & 0.400 \end{vmatrix} = -200(0.400)\hat{\mathbf{j}} + 200(0.300)\hat{\mathbf{k}}$$

Then, $$\vec{\mathbf{a}} = \frac{e}{m}\left[\vec{\mathbf{E}} + \vec{\mathbf{v}} \times \vec{\mathbf{B}}\right] = \left(\frac{1.60 \times 10^{-19}}{1.67 \times 10^{-27}}\right)\left[50.0\ \hat{\mathbf{j}} - 80.0\ \hat{\mathbf{j}} + 60.0\ \hat{\mathbf{k}}\right] \text{ m/s}^2$$

$$= \boxed{\left(-2.87 \times 10^{9}\hat{\mathbf{j}} + 5.75 \times 10^{9}\hat{\mathbf{k}}\right) \text{ m/s}^2}$$

P34.6 (a) The very long rod creates the same electric field that it would if stationary. We apply Gauss's law to a cylinder, concentric with the rod, of radius r = 20.0 cm and length ℓ:



ANS. FIG. P34.6

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{inside}}}{\epsilon_0}$$

$$E(2\pi r\ell)\cos 0° = \frac{\lambda \ell}{\epsilon_0}$$

$$\vec{\mathbf{E}} = \frac{\lambda}{2\pi\epsilon_0 r} \text{ radially outward}$$

$$= \frac{35.0 \times 10^{-9} \text{ C/m}}{2\pi\left(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2\right)(0.200 \text{ m})}\hat{\mathbf{j}}$$

$$= \boxed{3.15 \times 10^{3}\hat{\mathbf{j}} \text{ N/C}}$$

(b) The charge in motion constitutes a current of

$$(35.0 \times 10^{-9} \text{ C/m}) \times (15.0 \times 10^{6} \text{ m/s}) = 0.525 \text{ A}$$

This current creates a magnetic field.

$$\vec{\mathbf{B}} = \frac{\mu_0 I}{2\pi r}$$

$$= \frac{\left(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}\right)(0.525 \text{ A})}{2\pi(0.200 \text{ m})}\hat{\mathbf{k}} = \boxed{5.25\hat{\mathbf{k}} \times 10^{-7} \text{ T}}$$

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(c) The Lorentz force on the electron is $\vec{\mathbf{F}} = q\vec{\mathbf{E}} + q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$.

$$\vec{\mathbf{F}} = \left(-1.60 \times 10^{-19}\ \text{C}\right)\left(3.15 \times 10^{3}\,\hat{\mathbf{j}}\ \text{N/C}\right) + \left(-1.60 \times 10^{-19}\ \text{C}\right)\left(240 \times 10^{6}\,\hat{\mathbf{i}}\ \text{m/s}\right) \times \left(5.25 \times 10^{-7}\,\hat{\mathbf{k}}\ \text{T}\right)$$

$$\vec{\mathbf{F}} = 5.04 \times 10^{-16}\left(-\hat{\mathbf{j}}\right)\ \text{N} + 2.02 \times 10^{-17}\left(+\hat{\mathbf{j}}\right)\ \text{N}$$

$$= \boxed{4.83\left(-\hat{\mathbf{j}}\right) \times 10^{-16}\ \text{N}}$$

Section 34.3 Plane Electromagnetic Waves

***P34.7** (a) From Equation 34.20,

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}\ \text{m/s}}{1\ 150 \times 10^{3}\ \text{s}^{-1}} = 261\ \text{m}$$

so

$$\frac{180\ \text{m}}{261\ \text{m}} = \boxed{0.690\ \text{wavelengths}}$$

(b) From Equation 34.20,

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}\ \text{m/s}}{98.1 \times 10^{6}\ \text{s}^{-1}} = 3.06\ \text{m}$$

so

$$\frac{180\ \text{m}}{3.06\ \text{m}} = \boxed{58.9\ \text{wavelengths}}$$

***P34.8** From Equation 34.20,

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}\ \text{m/s}}{27.33 \times 10^{6}\ \text{Hz}} = \boxed{11.0\ \text{m}}$$

P34.9 (a) Since the light from this star travels at 3.00×10^{8} m/s, the last bit of light will hit the Earth in

$$t = \frac{d}{c} = \frac{6.44 \times 10^{18}\ \text{m}}{2.998 \times 10^{8}\ \text{m/s}} = 2.15 \times 10^{10}\ \text{s} = \boxed{681\ \text{years}}$$

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(b) From Table C.4 (in Appendix C of the textbook), the average Earth-Sun distance is $d = 1.496 \times 10^{11}$ m, giving the transit time as

$$t = \frac{d}{c} = \left(\frac{1.496 \times 10^{11} \text{ m}}{2.998 \times 10^{8} \text{ m/s}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right) = \boxed{8.32 \text{ min}}$$

(c) Also from Table C.4, the average Earth-Moon distance is $d = 3.84 \times 10^{8}$ m, giving the time for the round trip as

$$t = \frac{2d}{c} = \frac{2\left(3.84 \times 10^{8} \text{ m}\right)}{2.998 \times 10^{8} \text{ m/s}} = \boxed{2.56 \text{ s}}$$

P34.10 From $f\lambda = c$, we have

$$f = \frac{c}{\lambda} = \frac{2.998 \times 10^{8} \text{ m/s}}{632.8 \times 10^{-9} \text{ m}} = \boxed{4.738 \times 10^{14} \text{ Hz}}$$

P34.11 In the fundamental mode, there is a single loop in the standing wave between the plates. Therefore, the distance between the plates is equal to half a wavelength.

$$\lambda = 2L = 2(2.00 \text{ m}) = 4.00 \text{ m}$$

Thus, $$f = \frac{c}{\lambda} = \frac{3.00 \times 10^{8} \text{ m/s}}{4.00 \text{ m}} = 7.50 \times 10^{7} \text{ Hz} = \boxed{75.0 \text{ MHz}}.$$

P34.12 $$\frac{E}{B} = c \quad \text{or} \quad \frac{220 \text{ V/m}}{B} = 3.00 \times 10^{8} \text{ m/s, so}$$

$$B = 7.33 \times 10^{-7} \text{ T} = \boxed{733 \text{ nT}}$$

P34.13 From Equation 34.17,

$$v = \frac{1}{\sqrt{\kappa \mu_0 \epsilon_0}} = \frac{1}{\sqrt{1.78}} c = 0.750c = \boxed{2.25 \times 10^{8} \text{ m/s}}$$

***P34.14** Time to reach object

$$= \frac{1}{2}(\text{total time of flight}) = \frac{1}{2}\left(4.00 \times 10^{-4} \text{ s}\right) = 2.00 \times 10^{-4} \text{ s}$$

Thus,

$$d = vt = \left(3.00 \times 10^{8} \text{ m/s}\right)\left(2.00 \times 10^{-4} \text{ s}\right) = 6.00 \times 10^{4} \text{ m} = \boxed{60.0 \text{ km}}$$

P34.15 (a) $c = f\lambda$ gives the frequency as

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^{8} \text{ m/s}}{50.0 \text{ m}} = \boxed{6.00 \times 10^{6} \text{ Hz}}$$

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(b) $c = E/B$ gives the magnetic field amplitude as

$$B = \frac{E}{c} = \frac{22.0\ \text{V/m}}{3.00\times10^{8}\ \text{m/s}} = 7.33\times10^{-8}\ \text{T} = 73.3\ \text{nT}$$

$\vec{\mathbf{B}}$ must be directed along **negative** z **direction** when $\vec{\mathbf{E}}$ is in the negative y direction, so that $\vec{\mathbf{S}} = \vec{\mathbf{E}}\times\vec{\mathbf{B}}/\mu_0$ will propagate in the direction $\left(-\hat{\mathbf{j}}\right)\times\left(-\hat{\mathbf{k}}\right) = +\hat{\mathbf{i}}$. So,

$$\vec{\mathbf{B}}_{\max} = \boxed{-73.3\hat{\mathbf{k}}\ \text{nT}}$$

(c) $$k = \frac{2\pi}{\lambda} = \frac{2\pi}{50.0\,\text{m}} = 0.126\ \text{m}^{-1}$$

and $\omega = 2\pi f = 2\pi\left(6.00\times10^{6}\ \text{s}^{-1}\right) = 3.77\times10^{7}\ \text{rad/s}.$

Then,

$$\vec{\mathbf{B}} = \vec{\mathbf{B}}_{\max}\cos\left(kx-\omega t\right) = \boxed{-73.3\hat{\mathbf{k}}\cos\left(0.126x - 3.77\times10^{7}t\right)\ \text{nT}}$$

P34.16 $E = E_{\max}\cos\left(kx-\omega t\right)$

$$\frac{\partial E}{\partial x} = -E_{\max}\sin\left(kx-\omega t\right)(k) \quad\rightarrow\quad \frac{\partial^2 E}{\partial x^2} = -E_{\max}\cos\left(kx-\omega t\right)\left(k^2\right)$$

$$\frac{\partial E}{\partial t} = -E_{\max}\sin\left(kx-\omega t\right)(-\omega) \quad\rightarrow\quad \frac{\partial^2 E}{\partial t^2} = -E_{\max}\cos\left(kx-\omega t\right)(-\omega)^2$$

We must show: $\dfrac{\partial^2 E}{\partial x^2} = \mu_0 \in_0 \dfrac{\partial^2 E}{\partial t^2}$

That is, $-\left(k^2\right)E_{\max}\cos\left(kx-\omega t\right) = -\mu_0\in_0(-\omega)^2 E_{\max}\cos\left(kx-\omega t\right).$

But this is true, because $\dfrac{k^2}{\omega^2} = \left(\dfrac{1}{f\lambda}\right)^2 = \dfrac{1}{c^2} = \mu_0\in_0$.

The proof for the wave of the magnetic field follows precisely the same steps.

P34.17 Since the separation of the burn marks is $d_{\text{A to A}} = 6\ \text{cm}\pm5\% = \dfrac{\lambda}{2}$, then

$\lambda = 12\ \text{cm}\ \pm5\%$ and

$$v = \lambda f = (0.12\ \text{m}\ \pm5\%)\left(2.45\times10^{9}\ \text{s}^{-1}\right)$$

$$= \boxed{2.9\times10^{8}\ \text{m/s}\ \pm5\%}$$

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P34.18 The amplitudes of the electric and magnetic fields are in the correct ratio so that $E_{max}/B_{max} = c$. The ratio of ω to k, however, must also equal the speed of light:

$$\frac{\omega}{k} = \frac{3.00 \times 10^{15}\ \text{s}^{-1}}{9.00 \times 10^{6}\ \text{m}^{-1}} = 3.33 \times 10^{8}\ \text{m/s}$$

This value is higher than the speed of light in a vacuum, so the wave as described is impossible.

P34.19 The wave is of the form $E_y = E_{max} \sin(kx - \omega t)$.

(a) 100 V/m is the amplitude of the electric field, so the amplitude of the magnetic field is

$$B_{max} = \frac{E_{max}}{c} = \frac{100\ \text{V/m}}{3.00 \times 10^{8}\ \text{m/s}} = 3.33 \times 10^{-7}\ \text{T} = \boxed{0.333\ \mu\text{T}}$$

(b) We compare the given wave function with $y = A \sin(kx - \omega t)$ to see that the wave number is $k = 1.00 \times 10^{7}\,\text{m}^{-1}$. With $k = 2\pi/\lambda$, we then have the wavelength as

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{1.00 \times 10^{7}\ \text{m}^{-1}} = \boxed{0.628\ \mu\text{m}}$$

(c) The frequency is

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^{8}\ \text{m/s}}{6.28 \times 10^{-7}\ \text{m}} = \boxed{4.77 \times 10^{14}\ \text{Hz}}$$

Section 34.4 Energy Carried by Electromagnetic Waves

P34.20 From Equation 17.7, we recall that the intensity I a distance r from a point or spherical source is inversely proportional to the square of the distance: $I = P/4\pi r^2$. At the Earth, $r_1 = 1.496 \times 10^{11}$ m, the intensity is $I_1 = I_E$, then at distance r_2, the intensity $I_2 = 3I_E$. Then,

$$\frac{I_1}{I_2} = \left(\frac{r_2}{r_1}\right)^2$$

and

$$r_2 = r_1\sqrt{\frac{I_1}{I_2}} = \left(1.496 \times 10^{11}\ \text{m}\right)\sqrt{\frac{1}{3}} = \boxed{8.64 \times 10^{10}\ \text{m}}$$

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P34.21 In time interval Δt, sunlight travels distance $\Delta x = c\Delta t$. The intensity of the sunlight passing into a volume $\Delta V = A\Delta x$ in time Δt is

$$S = I = \frac{U}{A\Delta t} = \frac{U}{A\,\Delta x/c} = \frac{Uc}{V} = uc$$

$$\frac{\text{Energy}}{\text{Unit Volume}} = u = \frac{I}{c} = \frac{1\,000\ \text{W/m}^2}{3.00\times10^8\ \text{m/s}} = \boxed{3.33\ \mu\text{J/m}^3}$$

P34.22 (a) $$\frac{P}{\text{area}} = \frac{\text{energy}}{\Delta t\cdot\text{area}} = \frac{600\times10^3\ \text{Wh}}{(30\ \text{d})(13.0\ \text{m})(9.50\ \text{m})}\left(\frac{1\ \text{d}}{24\ \text{h}}\right) = \boxed{6.75\ \text{W/m}^2}$$

(b) The car uses gasoline at the rate of $(55\ \text{mi/h})\left(\frac{\text{gal}}{25\ \text{mi}}\right)$. Its rate of energy conversion is

$$P = 44.0\times10^6\ \text{J/kg}\left(\frac{2.54\ \text{kg}}{1\ \text{gal}}\right)(55\ \text{mi/h})\left(\frac{\text{gal}}{25\ \text{mi}}\right)\left(\frac{1\ \text{h}}{3\,600\ \text{s}}\right)$$

$$= 6.83\times10^4\ \text{W}$$

Its power-per-footprint-area is

$$\frac{P}{\text{area}} = \frac{6.83\times10^4\ \text{W}}{(2.10\ \text{m})(4.90\ \text{m})} = \boxed{6.64\times10^3\ \text{W/m}^2}$$

(c) A powerful automobile that is running on sunlight would have to carry on its roof a solar panel huge compared with the size of the car.

(d) Agriculture and forestry for food and fuels, space heating of large and small buildings, water heating, and heating for drying and many other processes are current and potential applications of solar energy.

P34.23 Power output = (power input)(efficiency).

Thus, $$\text{Power input} = \frac{\text{Power out}}{\text{eff}} = \frac{1.00\times10^6\ \text{W}}{0.300} = 3.33\times10^6\ \text{W}$$

and $$A = \frac{P}{I} = \frac{3.33\times10^6\ \text{W}}{1.00\times10^3\ \text{W/m}^2} = \boxed{3.33\times10^3\ \text{m}^2}$$

P34.24 (a) $$\vec{\mathbf{E}}\cdot\vec{\mathbf{B}} = \left(80.0\hat{\mathbf{i}} + 32.0\hat{\mathbf{j}} - 64.0\hat{\mathbf{k}}\right)(\text{N/C})\cdot\left(0.200\hat{\mathbf{i}} + 0.080\,0\hat{\mathbf{j}} + 0.290\hat{\mathbf{k}}\right)\ \mu\text{T}$$

$$\vec{\mathbf{E}}\cdot\vec{\mathbf{B}} = (16.0 + 2.56 - 18.56)\ \mu\text{T}\cdot\text{N/C}^2 = \boxed{0}$$

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(b) $\vec{\mathbf{S}} = \frac{1}{\mu_0}\vec{\mathbf{E}} \times \vec{\mathbf{B}}$

$$= \left(\frac{1}{4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}}\right)\left[\left(80.0\hat{\mathbf{i}} + 32.0\hat{\mathbf{j}} - 64.0\hat{\mathbf{k}}\right)\ \mathrm{N/C}\right] \times \left[\left(0.200\hat{\mathbf{i}} + 0.080\,0\hat{\mathbf{j}} + 0.290\hat{\mathbf{k}}\right)\ \mu\mathrm{T}\right]$$

$$\vec{\mathbf{S}} = \frac{\left(6.40\hat{\mathbf{k}} - 23.2\hat{\mathbf{j}} - 6.40\hat{\mathbf{k}} + 9.28\hat{\mathbf{i}} - 12.8\hat{\mathbf{j}} + 5.12\hat{\mathbf{i}}\right) \times 10^{-6}\ \mathrm{W/m^2}}{4\pi \times 10^{-7}}$$

$$\vec{\mathbf{S}} = \boxed{\left(11.5\hat{\mathbf{i}} - 28.6\hat{\mathbf{j}}\right)\ \mathrm{W/m^2}}$$

$= 30.9\ \mathrm{W/m^2}$ at $-68.1°$ from the $+x$ axis

P34.25 (a) $I = \frac{E_{\max}^2}{2\mu_0 c} = \frac{\left(3.00 \times 10^6\ \mathrm{V/m}\right)^2}{2\left(4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}\right)\left(3.00 \times 10^8\ \mathrm{m/s}\right)}$

$$I = \boxed{1.19 \times 10^{10}\ \mathrm{W/m^2}}$$

(b) $P = IA = I\pi r^2 = \left(1.19 \times 10^{10}\ \mathrm{W/m^2}\right)\pi\left(\frac{5.00 \times 10^{-3}\ \mathrm{m}}{2}\right)^2$

$$= \boxed{2.34 \times 10^5\ \mathrm{W}}$$

P34.26 The energy put into the water in each container by electromagnetic radiation can be written as $\Delta E = eP\Delta t = eIA\Delta t$, where e is the percentage absorption efficiency. This energy has the same effect as heat in raising the temperature of the water:

$$eIA\Delta t = mc\Delta T = \rho V c \Delta T$$

$$\Delta T = \frac{eI\ell^2\Delta t}{\rho \ell^3 c} = \frac{eI\Delta t}{\rho \ell c}$$

where ℓ is the edge dimension of the container and c the specific heat of water. For the small container,

$$\Delta T = \frac{0.700\left(25.0 \times 10^3\ \mathrm{W/m^2}\right)(480\ \mathrm{s})}{\left(10^3\ \mathrm{kg/m^3}\right)(0.060\ 0\ \mathrm{m})\left(4\,186\ \mathrm{J/kg \cdot °C}\right)} = \boxed{33.4°\mathrm{C}}$$

For the larger,

$$\Delta T = \frac{0.910\left(25.0 \times 10^3\ \mathrm{W/m^2}\right)(480\ \mathrm{s})}{\left(10^3\ \mathrm{kg/m^3}\right)(0.120\ \mathrm{m})\left(4\,186\ \mathrm{J/°C}\right)} = \boxed{21.7°\mathrm{C}}$$

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P34.27 (a) $B_{\max} = \dfrac{E_{\max}}{c}:\ B_{\max} = \dfrac{7.00\times10^{5}\ \text{N/C}}{3.00\times10^{8}\ \text{m/s}} = \boxed{2.33\ \text{mT}}$

(b) $I = \dfrac{E_{\max}^2}{2\mu_0 c}:$

$$I = \frac{\left(7.00\times10^{5}\ \text{V/m}\right)^2}{2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^{8}\ \text{m/s}\right)} = 6.50\times10^{8}\ \text{W/m}^2$$

$$= \boxed{650\ \text{MW/m}^2}$$

(c) $I = \dfrac{P}{A}:\quad P = IA = \left(6.50\times10^{8}\ \text{W/m}^2\right)\left[\dfrac{\pi}{4}\left(1.00\times10^{-3}\ \text{m}\right)^2\right] = \boxed{511\ \text{W}}$

P34.28 (a) We assume that the starlight moves through space without any of it being absorbed. The radial distance is

$$20\ \text{ly} = 20c\left(1\ \text{yr}\right) = 20\left(3.00\times10^{8}\ \text{m/s}\right)\left(3.16\times10^{7}\ \text{s}\right)$$

$$= 1.89\times10^{17}\ \text{m}$$

$$I = \frac{P}{4\pi r^2} = \frac{4.00\times10^{28}\ \text{W}}{4\pi\left(1.89\times10^{17}\ \text{m}\right)^2} = 8.88\times10^{-8}\ \text{W/m}^2$$

$$= \boxed{88.8\ \text{nW/m}^2}$$

(b) The Earth presents the projected target area of a flat circle:

$$P = IA = \left(8.88\times10^{-8}\ \text{W/m}^2\right)\left[\pi\left(6.37\times10^{6}\ \text{m}\right)^2\right]$$

$$= 1.13\times10^{7}\ \text{W} = \boxed{11.3\ \text{MW}}$$

P34.29 The Poynting vector is

$$S_{\text{avg}} = \frac{Power}{A} = \frac{Power}{4\pi r^2}.$$

In meters,

$$r = \left(5.00\ \text{mi}\right)\left(1\,609\ \text{m/mi}\right) = 8.04\times10^{3}\ \text{m}$$

and the intensity of the wave is

$$S = \frac{250\times10^{3}\ \text{W}}{4\pi(8\,045\ \text{m})^2} = \boxed{307\ \mu\text{W/m}^2}$$

P34.30 (a) The intensity of the broadcast waves is

$$I = \frac{B_{\max}^2 c}{2\mu_0} = \frac{P}{4\pi r^2}$$

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solving,

$$B_{\max} = \sqrt{\left(\frac{P}{4\pi r^2}\right)\left(\frac{2\mu_0}{c}\right)} = \sqrt{\left(\frac{P}{2\pi r^2}\right)\left(\frac{\mu_0}{c}\right)}$$

$$= \sqrt{\frac{\left(10.0\times10^3\ \text{W}\right)\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)}{2\pi\left(5.00\times10^3\ \text{m}\right)^2\left(3.00\times10^8\ \text{m/s}\right)}} = \boxed{5.16\times10^{-10}\ \text{T}}$$

(b) Since the magnetic field of the Earth is approximately 5×10^{-5} T, the Earth's field is some 100 000 times stronger.

P34.31 The average Poynting flux is

$$S_{\text{avg}} = \frac{P_{\text{avg}}}{4\pi r^2} = \frac{E_{\max}^2}{2\mu_0 c}$$

solving,

$$E_{\max} = \sqrt{2\mu_0 c S_{\text{avg}}} = \sqrt{\mu_0 c \frac{P_{\text{avg}}}{2\pi r^2}}$$

$$= \sqrt{\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^8\ \text{m/s}\right)\frac{4.00\times10^3\ \text{W}}{2\pi\left[4.00(1\ 609\ \text{m})\right]^2}}$$

$$= 0.076\,1\ \text{V/m}$$

The maximum emf (amplitude) induced in a length L of wire is

$$\Delta V_{\max} = E_{\max} L = \left(76.1\ \text{mV/m}\right)\left(0.650\ \text{m}\right) = \boxed{49.5\ \text{mV}}$$

***P34.32** Power $= SA = \dfrac{E_{\max}^2}{2\mu_0 c}\left(4\pi r^2\right)$

Solving for r,

$$r = \sqrt{\frac{P\mu_0 c}{2\pi E_{\max}^2}} = \sqrt{\frac{(100\ \text{W})\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^8\ \text{m/s}\right)}{2\pi(15.0\ \text{V/m})^2}}$$

$$= \boxed{5.16\ \text{m}}$$

***P34.33** (a) $P = I^2 R = 150\ \text{W}$

$$A = 2\pi rL = 2\pi\left(0.900\times10^{-3}\ \text{m}\right)\left(0.080\,0\ \text{m}\right) = 4.52\times10^{-4}\ \text{m}^2$$

$$S = \frac{P}{A} = \frac{150\ \text{W}}{4.52\times10^{-4}\ \text{m}^2} = \boxed{332\ \text{kW/m}^2}\ \text{(points radially inward)}$$

(b) $$B = \frac{\mu_0 I}{2\pi r} = \frac{(1.00\ \text{A})\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)}{2\pi\left(0.900\times10^{-3}\ \text{m}\right)} = \boxed{222\ \mu\text{T}}$$

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$$E = \frac{\Delta V}{\Delta x} = \frac{IR}{L} = \frac{150\ \text{V}}{0.080\ 0\ \text{m}} = \boxed{1.88\ \text{kV/m}}$$

Note that these values yield $S = \dfrac{EB}{\mu_0} = 332\ \text{kW/m}^2$, in agreement with the result from part (a).

P34.34 (a) $E_{\text{rms}} = cB_{\text{rms}} = \left(3.00\times10^{8}\ \text{m/s}\right)\left(1.80\times10^{-6}\ \text{T}\right) = \boxed{540\ \text{V/m}}$

(b) From Equation 34.25,

$$u_{\text{avg}} = \frac{\left(B_{\max}\right)^2}{2\mu_0} = \frac{\left(B_{\text{rms}}\right)^2}{\mu_0} = \frac{\left(1.80\times10^{-6}\ \text{T}\right)^2}{4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}} = \boxed{2.58\ \mu\text{J/m}^3}$$

(c) $S_{\text{avg}} = cu_{\text{avg}} = \left(3.00\times10^{8}\ \text{m/s}\right)\left(2.58\times10^{-6}\ \text{J/m}^3\right) = \boxed{773\ \text{W/m}^2}$

Section 34.5 Momentum and Radiation Pressure

P34.35 The intensity of the beam is $I = \dfrac{P_{\text{power}}}{\pi r^2}$, where $r = 1.00 \times 10^{-3}$ m. By Equation 34.29, the radiation pressure on the mirror is

$$P = \frac{2S}{c} = \frac{2I}{c} = \frac{2P_{\text{power}}}{\pi r^2 c}$$

$$= \frac{2\left(25.0\times10^{-3}\ \text{W}\right)}{\pi\left(1.00\times10^{-3}\ \text{m}\right)^2\left(3.00\times10^{8}\ \text{m/s}\right)} = \boxed{5.31\times10^{-5}\ \text{N/m}^2}$$

P34.36 For complete absorption, from equation 34.27,

$$P = \frac{S}{c} = \frac{25.0\ \text{W/m}^2}{3.00\times10^{8}\ \text{m/s}} = 8.33\times10^{-8}\ \text{N/m}^2 = \boxed{83.3\ \text{nPa}}$$

P34.37 (a) $I = \dfrac{P}{\pi r^2} = \dfrac{E_{\max}^2}{2\mu_0 c}$, and $r = 1.00 \times 10^{-3}$ m:

$$E_{\max} = \sqrt{\frac{2\mu_0 cP}{\pi r^2}}$$

$$= \sqrt{\frac{2\left[4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right]\left(3.00\times10^{8}\ \text{m/s}\right)\left(15.0\times10^{-3}\ \text{W}\right)}{\pi\left(1.00\times10^{-3}\ \text{m}\right)^2}}$$

$$= 1.90\times10^{8}\ \text{J} = \boxed{1.90\ \text{kN/C}}$$

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(b) The beam carries power P. The amount of energy ΔE in the length of a beam of length ℓ is the amount of power that passes a point in time interval $\Delta t = \ell/c$:

$$P = \frac{\Delta E}{\Delta t} = \frac{\Delta E}{\ell/c}$$

or $$\Delta E = \frac{P\ell}{c} = \frac{15.0\times10^{-3}\ \text{W}}{3.00\times10^{8}\ \text{m/s}}(1.00\ \text{m}) = \boxed{50.0\ \text{pJ}}.$$

(c) From Equation 34.27 and our result in part (b), the momentum and energy carried a light beam are related by

$$p = \frac{T_{ER}}{c} = \frac{\Delta E}{c} = \frac{50.0\times10^{-12}\ \text{J}}{3.00\times10^{8}\ \text{m/s}} = \boxed{1.67\times10^{-19}\ \text{kg}\cdot\text{m/s}}$$

P34.38 (a) $$I = \frac{P}{\pi r^2} = \frac{E_{max}^2}{2\mu_0 c} \rightarrow E_{max} = \boxed{\sqrt{\frac{2\mu_0 cP}{\pi r^2}}}$$

(b) The beam carries power P. The amount of energy ΔE in the length of a beam of length ℓ is the amount of power that passes a point in time interval $\Delta t = \ell/c$:

$$P = \frac{\Delta E}{\Delta t} = \frac{\Delta E}{\ell/c} \rightarrow \Delta E = \boxed{\frac{P\ell}{c}}$$

(c) From Equation 34.27 and our result in part (b), the momentum and energy carried a light beam are related by

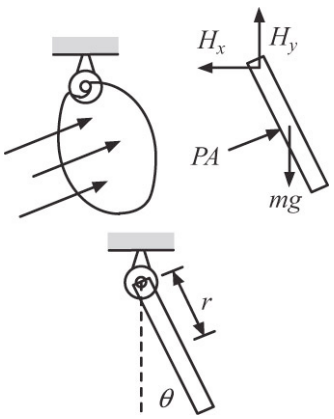
$$p = \frac{T_{ER}}{c} = \frac{\Delta E}{c} = \boxed{\frac{P\ell}{c^2}}$$

P34.39 The radiation pressure on the disk is

$$P = \frac{S}{c} = \frac{I}{c} = \frac{F}{A} = \frac{F}{\pi r^2}.$$

Thus, $$F = \frac{\pi r^2 I}{c}$$

Because the force acts uniformly over the surface of the disk, we may consider it to be acting at the center of the disk when calculating its torque. Take torques about the hinge:



ANS. FIG. P34.39

$\sum \tau = 0$:

$$H_x(0) + H_y(0) - mgr\sin\theta + \frac{\pi r^2 Ir}{c} = 0$$

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Solving for the angle gives

$$\theta = \sin^{-1}\left(\frac{\pi r^2 I}{mgc}\right)$$

$$= \sin^{-1}\left[\frac{\pi(0.400\text{ m})^2\left(10.0\times10^6\text{ W/ m}^2\right)}{(0.024\,0\text{ kg})(9.80\text{ m/ s}^2)(3.00\times10^8\text{ m/s})}\right]$$

$$= \sin^{-1} 0.071\,2 = \boxed{4.09°}$$

P34.40 (a) The light pressure on the absorbing Earth is $P = \frac{S}{c} = \frac{I}{c}$.

The force is

$$F = PA = \frac{I}{c}(\pi R^2) = \frac{(1\,370\text{ W/m}^2)\pi(6.37\times10^6\text{ m})^2}{3.00\times10^8\text{ m/s}}$$

$$= \boxed{5.82\times10^8\text{ N}}$$

away from the Sun.

(b) The attractive gravitational force exerted on Earth by the Sun is

$$F_g = \frac{GM_S M_M}{r_M^2}$$

$$= \frac{\left(6.67\times10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2\right)\left(1.991\times10^{30}\text{ kg}\right)\left(5.98\times10^{24}\text{ kg}\right)}{\left(1.496\times10^{11}\text{ m}\right)^2}$$

$$= 3.55\times10^{22}\text{ N}$$

which is $\boxed{6.10\times10^{13}\text{ times stronger}}$ compared to the repulsive force in part (a).

P34.41 (a) The magnitude of the momentum transferred to the assumed totally reflecting surface in time interval Δt is (from Equation 34.29)

$$\Delta p = \frac{2T_{ER}}{c} = \frac{2SA\Delta t}{c}$$

Then the momentum transfer is

$$\Delta\vec{\mathbf{p}} = \frac{2\vec{\mathbf{S}}A\Delta t}{c} = \frac{2(6.00\,\hat{\mathbf{i}}\text{ W/m}^2)(40.0\times10^{-4}\text{ m}^2)(1.00\text{ s})}{3.00\times10^8\text{ m/s}}$$

$$\Delta\vec{\mathbf{p}} = \boxed{1.60\times10^{-10}\,\hat{\mathbf{i}}\text{ kg}\cdot\text{m/s each second}}$$

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(b) The force is

$$\vec{\mathbf{F}} = PA\,\hat{\mathbf{i}} = \frac{2SA}{c}\,\hat{\mathbf{i}} = \frac{2\left(6.00\ \text{W/m}^2\right)\left(40.0\times10^{-4}\ \text{m}^2\right)(1.00\ \text{s})}{3.00\times10^{8}\ \text{m/s}}$$

$$= \boxed{1.60\times10^{-10}\ \hat{\mathbf{i}}\ \text{N}}$$

(c) The answers are the same. Force is the time rate of momentum transfer.

P34.42 (a) If P_S is the total power radiated by the Sun, and r_E and r_M are the radii of the orbits of the planets Earth and Mars, then the intensities of the solar radiation at these planets are:

$$I_E = \frac{P_S}{4\pi r_E^2} \quad \text{and} \quad I_M = \frac{P_S}{4\pi r_M^2}$$

Thus,

$$I_M = I_E\left(\frac{r_E}{r_M}\right)^2 = \left(1\,370\ \text{W/m}^2\right)\left(\frac{1.496\times10^{11}\ \text{m}}{2.28\times10^{11}\ \text{m}}\right)^2$$

$$= \boxed{590\ \text{W/m}^2}$$

(b) Mars intercepts the power falling on its circular face:

$$P_M = I_M\left(\pi R_M^2\right) = \left(590\ \text{W/m}^2\right)\left[\pi\left(3.37\times10^{6}\ \text{m}\right)^2\right]$$

$$= \boxed{2.10\times10^{16}\ \text{W}}$$

(c) If Mars behaves as a perfect absorber, it feels pressure

$$P = \frac{S_M}{c} = \frac{I_M}{c},$$

so the light-pressure force is

$$F_L = PA = \frac{I_M}{c}\left(\pi R_M^2\right) = \frac{P_M}{c} = \frac{2.10\times10^{16}\ \text{W}}{3.00\times10^{8}\ \text{m/s}} = \boxed{7.01\times10^{7}\ \text{N}}$$

(d) Using our results from above, we have $F_L = I_M\dfrac{\pi R_M^2}{c}$ and $I_M = I_E\dfrac{r_E^2}{r_M^2}$, so the light-pressure force on Mars is $F_L = I_E\dfrac{r_E^2}{r_M^2}\dfrac{\pi R_M^2}{c}$.

The attractive gravitational force exerted on Mars by the Sun is $F_g = \dfrac{GM_SM_M}{r_M^2}$. Their ratio is

$$\frac{F_g}{F_L} = \frac{GM_SM_M}{\cancel{r_M^2}}\cdot\frac{1}{I_E}\frac{\cancel{r_M^2}}{r_E^2}\frac{c}{\pi R_M^2} = \left(\frac{cGM_S}{\pi I_E r_E^2}\right)\frac{M_M}{R_M^2}$$

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Suppressing units,

$$\frac{F_g}{F_L} = \left[\frac{\left(3.00\times10^{8}\right)\left(6.67\times10^{-11}\right)\left(1.991\times10^{30}\right)}{\pi\left(1\,370\right)\left(1.496\times10^{11}\right)^2}\right]\left(\frac{M_M}{R_M^2}\right)$$

$$\frac{F_g}{F_L} = \left(414\ \text{m}^2/\text{kg}\right)\frac{M_M}{R_M^2} = \left(414\ \text{m}^2/\text{kg}\right)\frac{\left(6.42\times10^{23}\ \text{kg}\right)}{\left(3.37\times10^{6}\ \text{m}\right)^2}$$

$$= 2.34\times10^{13}$$

The attractive gravitational force exerted on Mars by the Sun is $\boxed{\sim 10^{13}\ \text{times stronger}}$ than the repulsive light-pressure force of part (c).

(e) The expression for the ratio of the gravitational force to the light-pressure force for Earth is similar to that used in part (d) for Mars (replace M with E):

$$\frac{F_g}{F_L} = \left(414\ \text{m}^2/\text{kg}\right)\frac{M_E}{R_E^2} = \left(414\ \text{m}^2/\text{kg}\right)\frac{\left(5.98\times10^{24}\ \text{kg}\right)}{\left(6.37\times10^{6}\ \text{m}\right)^2}$$

$$= 6.10\times10^{13}$$

The values are similar for both planets because both the forces follow inverse-square laws. The force ratios are not identical for the two planets because of their different radii and masses.

P34.43 (a) The radiation pressure is

$$P = \frac{2S}{c} = \frac{2I}{c}$$

The force on area A is

$$F = PA = \frac{2\left(1\,370\ \text{W/m}^2\right)}{3.00\times10^{8}\ \text{m/s}}\left(6.00\times10^{5}\ \text{m}^2\right) = \boxed{5.48\ \text{N}}$$

(b) The acceleration is:

$$a = \frac{F}{m} = \frac{5.48\ \text{N}}{6\,000\ \text{kg}} = 9.13\times10^{-4}\ \text{m/s}^2$$

$$= \boxed{913\ \mu\text{m/s}^2\ \text{away from the Sun}}$$

(c) It will arrive at time t, where $d = \frac{1}{2}at^2$ or,

$$t = \sqrt{\frac{2d}{a}} = \sqrt{\frac{2\left(3.84\times10^{8}\ \text{m}\right)}{\left(9.13\times10^{-4}\ \text{m/s}^2\right)}} = 9.17\times10^{5}\ \text{s} = \boxed{10.6\ \text{days}}$$

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Section 34.6 Production of Electromagnetic Waves by an Antenna

P34.44 (a) The wavelength of an ELF wave of frequency 75.0 Hz is

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{75.0 \text{ Hz}} = 4.00 \times 10^6 \text{ m}$$

The length of a quarter-wavelength antenna would be

$$L = 1.00 \times 10^6 \text{ m} = \boxed{1.00 \times 10^3 \text{ km}}$$

or $$L = (1\,000 \text{ km})\left(\frac{0.621 \text{ mi}}{1.00 \text{ km}}\right) = \boxed{621 \text{ mi}}$$

(b) While the project may be theoretically possible, it is not very practical.

P34.45 (a) $$h = \frac{\lambda}{4} = \frac{c}{4f} = \frac{3.00 \times 10^8 \text{ m/s}}{4(560 \times 10^3 \text{ Hz})} = \boxed{134 \text{ m}}$$

(b) $$h = \frac{\lambda}{4} = \frac{c}{4f} = \frac{3.00 \times 10^8 \text{ m/s}}{4(1600 \times 10^3 \text{ Hz})} = \boxed{46.9 \text{ m}}$$

P34.46 (a) The magnetic field $\vec{\mathbf{B}} = \frac{1}{2}\mu_0 J_{\max} \cos(kx - \omega t)\hat{\mathbf{k}}$ applies for $x > 0$, since it describes a wave moving in the $\hat{\mathbf{i}}$ direction. The electric field direction must satisfy $\vec{\mathbf{S}} = \frac{1}{\mu_0}\vec{\mathbf{E}} \times \vec{\mathbf{B}}$ as $\hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{k}}$ so the direction of the electric field is $\hat{\mathbf{j}}$ when the cosine is positive. For its magnitude we have $E = cB$, so altogether we have

$$\boxed{\vec{\mathbf{E}} = \frac{1}{2}\mu_0 c J_{\max} \cos(kx - \omega t)\hat{\mathbf{j}}}.$$

(b) $$\vec{\mathbf{S}} = \frac{1}{\mu_0}\vec{\mathbf{E}} \times \vec{\mathbf{B}} = \frac{1}{\mu_0}\frac{1}{4}\mu_0^2 c J_{\max}^2 \cos^2(kx - \omega t)\hat{\mathbf{i}}$$

$$\boxed{\vec{\mathbf{S}} = \frac{1}{4}\mu_0 c J_{\max}^2 \cos^2(kx - \omega t)\hat{\mathbf{i}}}$$

(c) The intensity is the magnitude of the Poynting vector averaged over one or more cycles. The average of the cosine-squared function is $\frac{1}{2}$, so $\boxed{I = \frac{1}{8}\mu_0 c J_{\max}^2}$.

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(d) $$J_{\max} = \sqrt{\frac{8I}{\mu_0 c}} = \sqrt{\frac{8\left(570\ \text{W/m}^2\right)}{4\pi \times 10^{-7}\left(\text{Tm/A}\right)3\times 10^8\ \text{m/s}}} = \boxed{3.48\ \text{A/m}}$$

***P34.47** For the proton, Newton's second law gives

$$\sum F = ma: \qquad qvB \sin 90.0° = \frac{mv^2}{R}.$$

The period and frequency of the proton's circular motion are therefore:

$$T = \frac{2\pi R}{v} = \frac{2\pi m}{qB} = \frac{2\pi\left(1.67\times 10^{-27}\ \text{kg}\right)}{\left(1.60\times 10^{-19}\ \text{C}\right)(0.350\ \text{T})} = 1.87\times 10^{-7}\ \text{s}$$

and $$f = 5.34\times 10^6\ \text{Hz}.$$

The charge will radiate at this same frequency, with

$$\lambda = \frac{c}{f} = \frac{3.00\times 10^8\ \text{m/s}}{5.34\times 10^6\ \text{Hz}} = \boxed{56.2\ \text{m}}$$

P34.48 For the proton, $\sum F = ma$ yields

$$qvB\sin 90.0° = \frac{mv^2}{R} \ \rightarrow\ v = \frac{qBR}{m}$$

The period of the proton's circular motion is therefore:

$$T = \frac{2\pi R}{v} = \frac{2\pi m}{qB}$$

The frequency of the proton's motion is $f = \dfrac{1}{T}$

The charge will radiate electromagnetic waves at this frequency, with

$$\lambda = \frac{c}{f} = cT = \boxed{\frac{2\pi mc}{qB}}$$

P34.49 Refer to ANS. FIG. P34.49. For any wavelength:

(a) Constructive interference occurs when $d\cos\theta = n\lambda$ for some integer n.

$$\cos\theta = n\frac{\lambda}{d} = n\left(\frac{\lambda}{\lambda/2}\right) = 2n \qquad n = 0,\ \pm 1,\ \pm 2,\ \ldots$$

$\therefore$ strong signal @ $\theta = \cos^{-1} 0 = 90°,\ 270°$, or

along the perpendicular bisector of the line segment joining the antennas.

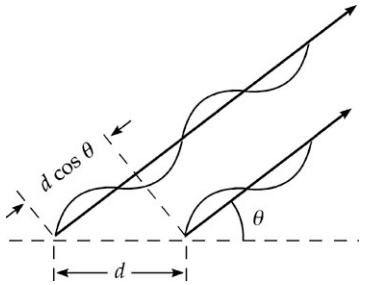
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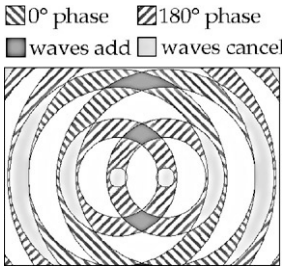
(b) Destructive interference occurs when

$$d\cos\theta = \left(\frac{2n+1}{2}\right)\lambda: \quad \cos\theta = 2n+1$$

$\therefore$ weak signal @ $\theta = \cos^{-1}(\pm 1) = 0°,\ 180°$, or

along the extensions of the line segment joining the antennas.





ANS. FIG. P34.49

Section 34.7 The Spectrum of Electromagnetic Waves

P34.50 (a) $f = \dfrac{c}{\lambda} = \dfrac{3\times10^8\ \text{m/s}}{1.7\ \text{m}}$ $\sim 10^8$ Hz radio wave

(b) 1 000 pages, 500 sheets, is about 3 cm thick so one sheet is about 6×10^{-5} m thick.

$$f = \frac{3.00\times10^8\ \text{m/s}}{6\times10^{-5}\ \text{m}}$$ $\sim 10^{13}$ Hz infrared

P34.51 (a) $f\lambda = c$ gives $\left(5.00\times10^{19}\ \text{Hz}\right)\lambda = 3.00\times10^8\ \text{m/s}$:

$\lambda = 6.00\times10^{-12}\ \text{m} = 6.00\ \text{pm}$

(b) $f\lambda = c$ gives $\left(4.00\times10^9\ \text{Hz}\right)\lambda = 3.00\times10^8\ \text{m/s}$:

$\lambda = 0.075\ 0\ \text{m} = 7.50\ \text{cm}$

P34.52 The time interval for the radio signal to travel 100 km is:

$$\Delta t_r = \frac{100\times10^3\ \text{m}}{3.00\times10^8\ \text{m/s}} = 3.33\times10^{-4}\ \text{s}$$

The sound wave travels 3.00 m across the room in:

$$\Delta t_s = \frac{3.00\ \text{m}}{343\ \text{m/s}} = 8.75\times10^{-3}\ \text{s}$$

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Therefore, listeners 100 km away will receive the news before the people in the newsroom by a total time difference of

$$\Delta t = 8.75 \times 10^{-3}\ \text{s} - 3.33 \times 10^{-4}\ \text{s} = 8.41 \times 10^{-3}\ \text{s}$$

P34.53 From $f\lambda = c$,

Channel 3: f = 60.0 MHz to 66.0 MHz.

(a) Channel 4: f = 66.0 MHz to 72.0 MHz, λ = 4.17 m to 4.55 m.

72.0–76.0 MHz is reserved for non-TV purposes.

Channel 5: f = 76.0 MHz to 82.0 MHz.

(b) Channel 6: f = 82.0 MHz to 88.0 MHz, λ = 3.41 m to 3.66 m.

88.0–174 MHz is reserved for non-TV purposes.

Channel 7: f = 174 MHz to 180 MHz.

(c) Channel 8: f = 180 MHz to 186 MHz, λ = 1.61 m to 1.67 m.

Additional Problems

***P34.54** From the electromagnetic spectrum chart and accompanying text discussion, the following identifications are made:

Frequency, f	Wavelength, $\lambda = \frac{c}{f}$	Classification
2 Hz = 2×10^{0} Hz	150 Mm	Radio
2 KHz = 2×10^{3} Hz	150 km	Radio
2 MHz = 2×10^{6} Hz	150 m	Radio
2 GHz = 2×10^{9} Hz	15 cm	Microwave
2 THz = 2×10^{12} Hz	150 μm	Infrared
2 PHz = 2×10^{15} Hz	150 nm	Ultraviolet
2 EHz = 2×10^{18} Hz	150 pm	X-ray
2 ZHz = 2×10^{21} Hz	150 fm	Gamma ray
2 YHz = 2×10^{24} Hz	150 am	Gamma ray

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Wavelength, λ	Frequency, $f = \frac{c}{\lambda}$	Classification
2 km = 2×10^3 m	1.5×10^5 Hz	Radio
2 m = 2×10^0 m	1.5×10^8 Hz	Radio
2 mm = 2×10^{-3} m	1.5×10^{11} Hz	Microwave
2 μm = 2×10^{-6} m	1.5×10^{14} Hz	Infrared
2 nm = 2×10^{-9} m	1.5×10^{17} Hz	Ultraviolet/X-ray
2 pm = 2×10^{-12} m	1.5×10^{20} Hz	X-ray/Gamma ray
2 fm = 2×10^{-15} m	1.5×10^{23} Hz	Gamma ray
2 am = 2×10^{-18} m	1.5×10^{26} Hz	Gamma ray

P34.55 (a) From $P = SA$, we have

$$P = \left(1\,370\ \text{W/m}^2\right)\left[4\pi\left(1.496\times10^{11}\ \text{m}\right)^2\right] = \boxed{3.85\times10^{26}\ \text{W}}$$

(b) $S = \dfrac{E_{\max}^2}{2\mu_0 c}$ so

$$E_{\max} = \sqrt{2\mu_0 cS}$$
$$= \sqrt{2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^8\ \text{m/s}\right)\left(1\,370\ \text{W/m}^2\right)}$$
$$= \boxed{1.02\ \text{kV/m}}$$

(c) $S = \dfrac{cB_{\max}^2}{2\mu_0}$ so

$$B_{\max} = \sqrt{\frac{2\mu_0 S}{c}} = \sqrt{\frac{2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(1\,370\ \text{W/m}^2\right)}{3.00\times10^8\ \text{m/s}}}$$
$$= \boxed{3.39\ \mu\text{T}}$$

P34.56 We use the relationship between energy density and electric field magnitude that we studied previously for a static field. The energy density can be written as

$$u_E = \tfrac{1}{2}\epsilon_0 E_{\max}^2$$

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so $$E_{\max} = \sqrt{\frac{2u_E}{\epsilon_0}} = \sqrt{\frac{2\left(4.00 \times 10^{-14}\ \text{N}\cdot\text{m}^2\right)}{8.85 \times 10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2}} = \boxed{95.1\ \text{mV/m}}.$$

P34.57 The wavelength is found from

$$f\lambda = c \ \rightarrow\ \lambda = \frac{c}{f} = \frac{3.00\times10^8\ \text{m/s}}{5.45\times10^{14}\ \text{Hz}} = \boxed{5.50\times10^{-7}\ \text{m}}$$

P34.58 The angular frequency of the wave is

$$\omega = 2\pi f = 2\pi\left(3.00\times10^9\ \text{s}^{-1}\right) = 1.88\times10^{10}\ \text{s}^{-1}$$

and the wave number is

$$k = \frac{2\pi}{\lambda} = \frac{\omega}{c} = 2\pi\left(\frac{3.00\times10^9\ \text{s}^{-1}}{3.00\times10^8\ \text{m/s}}\right) = 20.0\pi\ \text{m}^{-1} = 62.8\ \text{m}^{-1}$$

Also,

$$B_{\max} = \frac{E}{c} = \frac{300\ \text{V/m}}{3.00\times10^8\ \text{m/s}} = 1.00\ \mu\text{T}$$

Then,

$$\boxed{E = 300\cos\left(62.8x - 1.88\times10^{10}t\right)}$$

$$\boxed{B = 1.00\cos\left(62.8x - 1.88\times10^{10}t\right)}$$

where E is in volts per meter (V/m), B is in microtesla (μT), x is in meters, and t is in seconds.

***P34.59** (a) The power incident on the mirror is:

$$P_I = IA = \left(1\ 370\ \text{W/m}^2\right)\left[\pi(100\ \text{m})^2\right] = 4.30\times10^7\ \text{W}.$$

The power reflected through the atmosphere is

$$P_R = 0.746\left(4.30\times10^7\ \text{W}\right) = \boxed{3.21\times10^7\ \text{W}}$$

(b) $$S = \frac{P_R}{A} = \frac{3.21\times10^7\ \text{W}}{\pi\left(4.00\times10^3\ \text{m}\right)^2} = \boxed{0.639\ \text{W/m}^2}$$

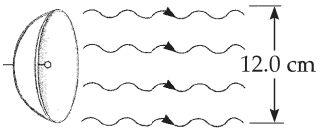
(c) Noon sunshine in St. Petersburg produces this power-per-area on a horizontal surface:

$$\frac{P_N}{A} = 0.746\left(1\ 370\ \text{W/m}^2\right)\sin 7.00^\circ = 125\ \text{W/m}^2$$

The radiation intensity received from the mirror is $\left(\frac{0.639\ \text{W/m}^2}{125\ \text{W/m}^2}\right)100\% = \boxed{0.513\%}$ of that from the noon Sun in January.

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***P34.60** (a) $\lambda = \dfrac{c}{f} = \dfrac{3.00\times10^{8}\ \text{m/s}}{20.0\times10^{9}\ \text{s}^{-1}} = \boxed{1.50\ \text{cm}}$



ANS. FIG. P34.60

(b) $U = P(\Delta t) = (25.0\times10^{3}\ \text{J/s})(1.00\times10^{-9}\ \text{s})$

$= 25.0\times10^{-6}\ \text{J} = \boxed{25.0\ \mu\text{J}}$

(c) $u_{\text{avg}} = \dfrac{U}{V} = \dfrac{U}{(\pi r^2)\ell}$

$= \dfrac{U}{(\pi r^2)c(\Delta t)} = \dfrac{25.0\times10^{-6}\ \text{J}}{\pi(0.060\ 0\ \text{m})^2(3.00\times10^{8}\ \text{m/s})(1.00\times10^{-9}\ \text{s})}$

$u_{\text{avg}} = 7.37\times10^{-3}\ \text{J/m}^3 = \boxed{7.37\ \text{mJ/m}^3}$

(d) $E_{\max} = \sqrt{\dfrac{2u_{\text{av}}}{\epsilon_0}} = \sqrt{\dfrac{2(7.37\times10^{-3}\ \text{J/m}^3)}{8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2}} = 4.08\times10^{4}\ \text{V/m}$

$= \boxed{40.8\ \text{kV/m}}$

$B_{\max} = \dfrac{E_{\max}}{c} = \dfrac{4.08\times10^{4}\ \text{V/m}}{3.00\times10^{8}\ \text{m/s}} = 1.36\times10^{-4}\ \text{T} = \boxed{136\ \mu\text{T}}$

(e) $F = PA = \left(\dfrac{S}{c}\right)A = u_{\text{av}}A = (7.37\times10^{-3}\ \text{J/m}^3)\pi(0.060\ 0\ \text{m})^2$

$= 8.33\times10^{-5}\ \text{N} = \boxed{83.3\ \mu\text{N}}$

P34.61 Suppose you cover a 1.7 m × 0.3 m section of beach blanket. Suppose the elevation angle of the Sun is 60°. Then the effective target area you fill in the Sun's light is

$$A = (1.7\ \text{m})(0.3\ \text{m})\cos 30^\circ = 0.4\ \text{m}^2$$

Now $I = \dfrac{P}{A} = \dfrac{\Delta E}{A\Delta t}$, so

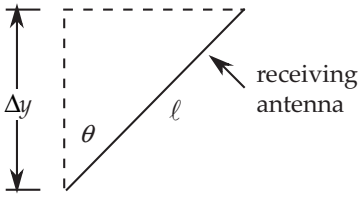
$$\Delta E = IA\Delta t = (0.5)\left[(0.6)(1\,370\ \text{W/m}^2)\right](0.4\ \text{m}^2)(3\,600\ \text{s})$$

$$\boxed{\sim 10^{6}\ \text{J}}$$

P34.62 $P = \dfrac{(\Delta V)^2}{R}$ or $P \propto (\Delta V)^2$

$\Delta V = (-)E_y \cdot \Delta y = E_y \cdot \ell\cos\theta$

$\Delta V \propto \cos\theta$ so $P \propto \cos^2\theta$



ANS. FIG. P34.62

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(a) $\theta = 15.0°$: $P = P_{\max}\cos^2(15.0°) = 0.933P_{\max} = \boxed{93.3\%}$

(b) $\theta = 45.0°$: $P = P_{\max}\cos^2(45.0°) = 0.500P_{\max} = \boxed{50.0\%}$

(c) $\theta = 90.0°$: $P = P_{\max}\cos^2(90.0°) = \boxed{0}$

P34.63 The gravitational force exerted by the Sun on the particle is given by

$$F_{\text{grav}} = \frac{GM_S m}{R^2} = \left(\frac{GM_S}{R^2}\right)\left[\rho\left(\frac{4}{3}\pi r^3\right)\right]$$

where M_S = mass of Sun, r = radius of particle, and R = distance from Sun to particle. The force exerted by solar radiation on the particle is given by $F_{\text{rad}} = PA,$ and since the particle absorbs all the radiation, by Equation 34.28, we have

$$F_{\text{rad}} = PA = \frac{S}{c}\pi r^2$$

When the particle is in equilibrium, the gravitational force toward the Sun is balanced by the force of radiation away from the Sun, $F_{\text{rad}} = F_{\text{grav}}$, so

$$\frac{S}{c}\pi r^2 = \left(\frac{GM_S}{R^2}\right)\left[\rho\left(\frac{4}{3}\pi r^3\right)\right]$$

Solving for r, the radius of the particle, then gives

$$r = \frac{3SR^2}{4cGM_S\rho}$$

Suppressing units,

$$r = \frac{3(214)\left(3.75\times10^{11}\right)^2}{4\left(3.00\times10^{8}\right)\left(6.67\times10^{-11}\right)\left(1.991\times10^{30}\right)(1\ 500)}$$

$$= 3.78\times10^{-7}\ \text{m} = \boxed{378\ \text{nm}}$$

P34.64 The gravitational force exerted by the Sun on the particle is given by

$$F_{\text{grav}} = \frac{GM_S m}{R^2} = \left(\frac{GM_S}{R^2}\right)\left[\rho\left(\frac{4}{3}\pi r^3\right)\right]$$

where M_S = mass of Sun, r = radius of particle, and R = distance from Sun to particle. The force exerted by solar radiation on the particle is given by $F_{\text{rad}} = PA,$ and since the particle absorbs all the radiation, by Equation 34.28, we have

$$F_{\text{rad}} = PA = \frac{S}{c}\pi r^2$$

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When the particle is in equilibrium, the gravitational force toward the Sun is balanced by the force of radiation away from the Sun, $F_{\text{rad}} = F_{\text{grav}}$, so

$$\frac{S}{c}\pi r^2 = \left(\frac{GM_S}{R^2}\right)\left[\rho\left(\frac{4}{3}\pi r^3\right)\right]$$

Solving for r, the radius of the particle, then gives

$$r = \boxed{\frac{3SR^2}{4cGM_S\rho}}$$

P34.65 (a) The magnetic-field amplitude is

$$B_{\max} = \frac{E_{\max}}{c} = \frac{0.200\times10^{-6}\ \text{V/m}}{3.00\times10^{8}\ \text{m/s}} = \boxed{6.67\times10^{-16}\ \text{T}}$$

(b) The intensity is the Poynting vector averaged over one or more cycles, given by

$$S_{\text{avg}} = \frac{E_{\max}^2}{2\mu_0 c} = \frac{\left(0.200\times10^{-6}\ \text{V/m}\right)^2}{2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}$$

$$= \boxed{5.31\times10^{-17}\ \text{W/m}^2}$$

(c) The power tells how fast the antenna receives energy. It is

$$P = S_{\text{avg}}A = S_{\text{avg}}\pi\left(\frac{d}{2}\right)^2 = \left(5.31\times10^{-17}\ \text{W/m}^2\right)\pi\left(\frac{20.0\ \text{m}}{2}\right)^2$$

$$= \boxed{1.67\times10^{-14}\ \text{W}}$$

(d) The force tells how fast the antenna receives momentum. It is

$$F = PA = \left(\frac{S_{\text{avg}}}{c}\right)A = \left(\frac{5.31\times10^{-17}\ \text{W/m}^2}{3.00\times10^{8}\ \text{m/s}}\right)\pi\left(\frac{20.0\ \text{m}}{2}\right)^2$$

$$= \boxed{5.56\times10^{-23}\ \text{N}}$$

(approximately the weight of 3 000 hydrogen atoms!)

P34.66 Of the intensity $S = 1\,370\ \text{W/m}^2$, the 38.0% that is reflected exerts a pressure

$$P_1 = \frac{2S_r}{c} = \frac{2(0.380)S}{c}$$

The absorbed light exerts pressure

$$P_2 = \frac{S_a}{c} = \frac{0.620S}{c}$$

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Altogether the pressure at the subsolar point on Earth is

(a) $$P_{\text{total}} = P_1 + P_2 = \frac{1.38S}{c} = \frac{1.38\left(1370\ \text{W/m}^2\right)}{3.00\times10^8\ \text{m/s}} = \boxed{6.30\times10^{-6}\ \text{Pa}}$$

(b) Compared to normal atmospheric pressure,

$$\frac{P_a}{P_{\text{total}}} = \frac{1.01\times10^5\ \text{N/m}^2}{6.30\times10^{-6}\ \text{N/m}^2}$$

$$= \boxed{1.60\times10^{10}\ \text{times smaller than atmospheric pressure}}$$

P34.67 The mirror intercepts power

$$P = I_1A_1 = \left(1.00\times10^3\ \text{W/m}^2\right)\left[\pi\left(0.500\ \text{m}\right)^2\right] = 785\ \text{W}.$$

(a) In the image, $I_2 = \dfrac{P}{A_2}$, so

$$I_2 = \frac{785\ \text{W}}{\pi\left(0.020\,0\ \text{m}\right)^2} = \boxed{625\ \text{kW/m}^2}$$

(b) $I_2 = \dfrac{E_{\max}^2}{2\mu_0 c}$, so

$$E_{\max} = \sqrt{2\mu_0 c I_2}$$

$$= \sqrt{2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^8\ \text{m/s}\right)\left(6.25\times10^5\ \text{W/m}^2\right)}$$

$$= \boxed{21.7\ \text{kN/C}}$$

(c) $$B_{\max} = \frac{E_{\max}}{c} = \boxed{72.4\ \mu\text{T}}$$

(d) We obtain the time interval from

$$0.400(P\Delta t) = mc\Delta T$$

solving,

$$\Delta t = \frac{mc\Delta T}{0.400P} = \frac{\left(1.00\ \text{kg}\right)\left(4\,186\ \text{J/kg}\cdot\text{°C}\right)(100\text{°C} - 20.0\text{°C})}{0.400(785\ \text{W})}$$

$$= 1.07\times10^3\ \text{s} = \boxed{17.8\ \text{min}}$$

P34.68 (a) In $\vec{\mathbf{E}} = \dfrac{q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}} = \dfrac{\Phi}{4\pi r^2}\hat{\mathbf{r}} = \dfrac{487\ \text{N}\cdot\text{m}^2/\text{C}}{4\pi r^2}\hat{\mathbf{r}}$,

$$\boxed{\vec{\mathbf{E}} = \frac{38.8}{r^2}\hat{\mathbf{r}} \text{ where } \vec{\mathbf{E}} \text{ is in volts per meter and } r \text{ is in meters.}}$$

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(b) The radiated intensity is

$$I = \frac{P}{4\pi r^2} = \frac{E_{\max}^2}{2\mu_0 c}$$

solving,

$$E_{\max} = \sqrt{\frac{2\mu_0 cP}{4\pi r^2}} = \frac{1}{r}\sqrt{\frac{\mu_0 cP}{2\pi}}$$

$$= \frac{1}{r}\sqrt{\frac{\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times 10^{8}\ \text{m/s}\right)(25.0\ \text{W})}{2\pi}}$$

$$\boxed{E_{\max} = \frac{38.7}{r} \text{ where } E \text{ is in volts per meter and } r \text{ is in meters.}}$$

(c) For $E_{\max} = \dfrac{38.7}{r} = 3.00\times 10^{6} \ \rightarrow \ r = 1.29\times 10^{-5} = 12.9\times 10^{-6}$, so r is 12.9 μm, but the expression in part (b) does not apply if this point is inside the source.

(d) From part (c), we see that in the radiated wave, the field amplitude is inversely proportional to distance. As the distance doubles, the amplitude is cut in half.

(e) In the static case, the field is inversely proportional to the square of the distance. As the distance doubles, the field is reduced by a factor of 4.

P34.69 (a) At steady state, $P_{\text{in}} = P_{\text{out}}$ and the power radiated out is $P_{\text{out}} = e\sigma AT^4$. Thus,

$$T = \left[\frac{P_{\text{out}}}{e\sigma A}\right]^{1/4} = \left[\frac{900\ \text{W/m}^2}{0.700\left(5.67\times 10^{-8}\ \text{W/m}^2\cdot\text{K}^4\right)}\right]^{1/4}$$

$$= \boxed{388\ \text{K}} = 115°\text{C}$$

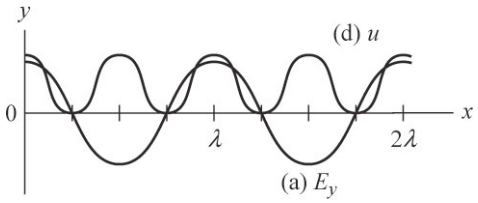
(b) The box of horizontal area A presents projected area $A \sin 50.0°$ perpendicular to the sunlight. Then by the same reasoning,

$$0.900\left(1\,000\ \text{W/m}^2\right)A\sin 50.0° = 0.700\left(5.67\times 10^{-8}\ \text{W/m}^2\cdot\text{K}^4\right)AT^4$$

or $$T = \left[\frac{\left(900\ \text{W/m}^2\right)\sin 50.0°}{0.700\left(5.67\times 10^{-8}\ \text{W/m}^2\cdot\text{K}^4\right)}\right]^{1/4} = \boxed{363\ \text{K}} = 90.0°\text{C}$$

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P34.70 (a) See ANS. FIG. P34.70



ANS. FIG. P34.70

(b) $$u_E = \frac{1}{2}\epsilon_0 E^2 = \boxed{\frac{1}{2}\epsilon_0 E_{\max}^2 \cos^2(kx)}$$

(c) $$u_B = \frac{1}{2\mu_0}B^2 = \boxed{\frac{1}{2\mu_0}B_{\max}^2 \cos^2(kx)}$$

(d) Note that

$$u_B = \frac{1}{2\mu_0}\frac{E_{\max}^2}{c^2}\cos^2(kx) = \frac{1}{2\mu_0}\frac{E_{\max}^2}{(1/\mu_0\epsilon_0)}\cos^2(kx)$$
$$= \frac{1}{2}\epsilon_0 E_{\max}^2 \cos^2(kx) = u_E$$

Therefore, $u = u_E + u_B = \boxed{\epsilon_0 E_{\max}^2 \cos^2(kx)}$.

(e) $$E_\lambda = \int_0^\lambda uA\,dx$$

$$E_\lambda = \int_0^\lambda \epsilon_0 E_{\max}^2 \cos^2(kx) A\,dx = \int_0^\lambda \epsilon_0 E_{\max}^2 A\left[\frac{1}{2} + \frac{1}{2}\cos(2kx)\right]A\,dx$$
$$= \frac{1}{2}\epsilon_0 E_{\max}^2 A\,x\Big|_0^\lambda + \frac{\epsilon_0 E_{\max}^2}{4k}A\sin(2kx)\Big|_0^\lambda$$
$$= \frac{1}{2}\epsilon_0 E_{\max}^2 A\lambda + \frac{\epsilon_0 E_{\max}^2}{4k}A[\sin(4\pi) - \sin(0)]$$
$$= \boxed{\frac{1}{2}\epsilon_0 E_{\max}^2 \lambda A}$$

(f) $$P = \frac{E_\lambda}{T} = \frac{1}{2}\frac{\epsilon_0 E_{\max}^2 \lambda A}{(1/f)} = \frac{1}{2}\epsilon_0 E_{\max}^2 (\lambda f)A = \boxed{\frac{1}{2}\epsilon_0 c E_{\max}^2 A}$$

(g) $$I = \frac{P}{A} = \frac{\frac{1}{2}\epsilon_0 c E_{\max}^2 A}{A} = \boxed{\frac{1}{2}\epsilon_0 c E_{\max}^2}$$

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(h) From part (g), we have

$$\frac{1}{2}\epsilon_0 cE_{\max}^2 = \frac{\mu_0}{\mu_0}\frac{\epsilon_0 cE_{\max}^2}{2} = (\mu_0 \epsilon_0)\frac{cE_{\max}^2}{2\mu_0} = \frac{1}{c^2}\frac{cE_{\max}^2}{2\mu_0} = \frac{E_{\max}^2}{2\mu_0 c}$$

The result in part (g) agrees with $I = \dfrac{E_{\max}^2}{2\mu_0 c}$ in Equation 34.24.

P34.71 The bead is black, so we assume it absorbs all light that strikes it. The bead presents an effective face of area $A = \pi r^2$ to the light. Since we assume the bead to be perfectly absorbing, the light pressure, from Equation 34.28, is

$$P = \frac{S_{\text{av}}}{c} = \frac{I}{c} = \frac{F_\ell}{A}$$

so the light force is $F_\ell = \dfrac{I}{c}A$.

(a) The light force balances the weight, $F_\ell = F_g$, so

$$\frac{I}{c}\pi r^2 = mg$$

solving,

$$I = \frac{mgc}{\pi r^2} = \frac{\rho\left(\frac{4}{3}\pi r^3\right)gc}{\pi r^2} = \frac{4}{3}\rho cgr$$

$$= \frac{4}{3}\left(\frac{0.200\times10^{-3}\ \text{kg}}{10^{-6}\ \text{m}^3}\right)\left(3.00\times10^8\ \text{m/s}\right)\left(9.80\ \text{m/s}^2\right)$$

$$\times\left(0.500\times10^{-3}\ \text{m}\right)$$

$$I = \boxed{3.92\times10^8\ \text{W/m}^2}$$

(b) The minimum power required is

$$P = IA = \left(3.92\times10^8\ \text{W/m}^2\right)\pi\left(0.500\times10^{-3}\ \text{m}\right)^2 = \boxed{308\ \text{W}}$$

P34.72 The bead is black, so we assume it absorbs all light that strikes it. The bead presents an effective face of area $A = \pi r^2$ to the light. Since we assume the bead to be perfectly absorbing, the light pressure, from Equation 34.28, is

$$P = \frac{S_{\text{av}}}{c} = \frac{I}{c} = \frac{F_\ell}{A}$$

so the light force is $F_\ell = \dfrac{I}{c}A$.

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(a) The light force balances the weight, $F_\ell = F_g$, so

$$\frac{I}{c}\pi r^2 = mg$$

solving,

$$I = \frac{mgc}{\pi r^2} = \frac{\rho\left(\frac{4}{3}\pi r^3\right)gc}{\pi r^2} = \boxed{\frac{4}{3}\rho cgr}$$

(b) The minimum power required is

$$P = IA = \left(\frac{4}{3}\rho cgr\right)\left(\pi r^2\right) = \boxed{\frac{4}{3}\pi\rho cgr^3}$$

P34.73 (a) A hemisphere is half a sphere:

$$m = \rho V = \rho\left[\frac{1}{2}\left(\frac{4}{3}\pi r^3\right)\right] = 5.50 + 4(0.800)\text{ kg} = 8.70\text{ kg}$$

$$r = \left(\frac{6m}{\rho 4\pi}\right)^{1/3} = \left(\frac{6(8.7\text{ kg})}{(990\text{ kg/m}^3)4\pi}\right)^{1/3} = \boxed{0.161\text{ m}}$$

(b) $A = \frac{1}{2}4\pi r^2 = 2\pi(0.161\text{ m})^2 = \boxed{0.163\text{ m}^2}$

(c) $P = e\sigma AT^4$ and $T = 31.0 + 273.0 = 304$ K:

$$P = 0.970\left(5.67\times10^{-8}\text{ W/m}^2\cdot\text{K}^4\right)(0.163\text{ m}^2)(304\text{ K})^4$$

$$= \boxed{76.8\text{ W}}$$

(d) $I = \frac{P}{A} = e\sigma T^4$

$$I = 0.970\left(5.67\times10^{-8}\text{ W/m}^2\cdot\text{K}^4\right)(304\text{ K})^4$$

$$= \boxed{470\text{ W/m}^2}$$

(e) $I = \frac{E_{\max}^2}{2\mu_0 c}$

$$E_{\max} = \left(2\mu_0 cI\right)^{1/2}$$

$$= \left[2\left(4\pi\times10^{-7}\text{ Tm/A}\right)\left(3.00\times10^8\text{ m/s}\right)\left(470\text{ W/m}^2\right)\right]^{1/2}$$

$$= \boxed{595\text{ N/C}}$$

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(f) $E_{\max} = cB_{\max} \rightarrow B_{\max} = \dfrac{595 \text{ N/C}}{3\times10^8 \text{ m/s}} = \boxed{1.98\ \mu\text{T}}$

(g) Each kitten has radius $r_k = \left(\dfrac{6m}{\rho 4\pi}\right)^{1/3} = \left[\dfrac{6(0.800)}{990\times 4\pi}\right]^{1/3} = 0.072\,8 \text{ m}$ and radiating area $2\pi(0.072\,8 \text{ m})^2 = 0.033\,3 \text{ m}^2$. The mother cat has area $2\pi\left[\dfrac{6(5.50)}{990\times 4\pi}\right]^{2/3} = 0.120 \text{ m}^2$. The total glowing area is $0.120 \text{ m}^2 + 4(0.033\,3 \text{ m}^2) = 0.254 \text{ m}^2$ and has power output $P = IA = (470 \text{ W/m}^2)(0.254 \text{ m}^2) = \boxed{119 \text{ W}}$.

P34.74 (a) On the right side of the equation,

$$\frac{\text{C}^2(\text{m/s}^2)^2}{(\text{C}^2/\text{N}\cdot\text{m}^2)(\text{m/s})^3} = \frac{\text{N}\cdot\text{m}^2\cdot\text{C}^2\cdot\text{m}^2\cdot\text{s}^3}{\text{C}^2\cdot\text{s}^4\cdot\text{m}^3} = \frac{\text{N}\cdot\text{m}}{\text{s}} = \frac{\text{J}}{\text{s}} = \text{W}$$

(b) $F = ma = qE$, or

$$a = \frac{qE}{m} = \frac{(1.60\times10^{-19} \text{ C})(100 \text{ N/C})}{9.11\times10^{-31} \text{ kg}} = \boxed{1.76\times10^{13} \text{ m/s}^2}$$

(c) The radiated power is then:

$$P = \frac{q^2a^2}{6\pi\epsilon_0 c^3} = \frac{(1.60\times10^{-19} \text{ C})^2(1.76\times10^{13} \text{ m/s}^2)^2}{6\pi(8.85\times10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2)(3.00\times10^8 \text{ m/s})^3}$$

$$= \boxed{1.75\times10^{-27} \text{ W}}$$

(d) $F = ma_c = m\left(\dfrac{v^2}{r}\right) = qvB$,

so $v = \dfrac{qBr}{m}$

The proton accelerates at

$$a = \frac{v^2}{r} = \frac{q^2B^2r}{m^2} = \frac{(1.60\times10^{-19} \text{ C})^2(0.350 \text{ T})^2(0.500 \text{ m})}{(1.67\times10^{-27} \text{ kg})^2}$$

$$= 5.62\times10^{14} \text{ m/s}^2$$

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The proton then radiates

$$P = \frac{q^2 a^2}{6\pi \epsilon_0 c^3} = \frac{\left(1.60\times10^{-19}\ \text{C}\right)^2\left(5.62\times10^{14}\ \text{m/s}^2\right)^2}{6\pi\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)\left(3.00\times10^{8}\ \text{m/s}\right)^3}$$

$$= \boxed{1.80\times10^{-24}\ \text{W}}$$

P34.75 We take R to be the planet's distance from its star, and r to be the radius of the planet.

(a) The effective area of the planet over which it absorbs light is its projection onto a plane perpendicular to the light from its sun. The projected area of a planet of radius r is πr^2, so the planet absorbs light over area πr^2.

(b) The planet radiates over its entire surface area, $4\pi r^2$.

(c) At steady-state, $P_{\text{in}} = P_{\text{out}}$:

$$eI_{\text{in}}\left(\pi r^2\right) = e\sigma\left(4\pi r^2\right)T^4$$

$$e\left(\frac{6.00\times10^{23}\ \text{W}}{4\pi R^2}\right)\left(\pi r^2\right) = e\sigma\left(4\pi r^2\right)T^4\text{, so that}$$

$$6.00\times10^{23}\ \text{W} = 16\pi\sigma R^2 T^4$$

$$R = \sqrt{\frac{6.00\times10^{23}\ \text{W}}{16\pi\sigma T^4}}$$

$$= \sqrt{\frac{6.00\times10^{23}\ \text{W}}{16\pi\left(5.67\times10^{-8}\ \text{W/m}^2\cdot\text{K}^4\right)(310\ \text{K})^4}} = \boxed{4.77\times10^{9}\ \text{m}}$$

Challenge Problems

P34.76 We are given $f = 90.0$ MHz and $E_{\max} = 200\ \text{mV/m} = 2.00\times10^{-3}\ \text{V/m}$

(a) The wavelength of the wave is $\lambda = \frac{c}{f} = \boxed{3.33\ \text{m}}$

(b) Its period is $T = \frac{1}{f} = 1.11\times10^{-8}\ \text{s} = \boxed{11.1\ \text{ns}}$

(c) We obtain the maximum value of the magnetic field from

$$B_{\max} = \frac{E_{\max}}{c} = 6.67\times10^{-12}\ \text{T} = \boxed{6.67\ \text{pT}}$$

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(d) $\boxed{\vec{\mathbf{E}} = \left(2.00\times10^{-3}\right)\cos 2\pi\left(\frac{x}{3.33} - 90.0\times10^{6}t\right)\hat{\mathbf{j}}}$

$\boxed{\vec{\mathbf{B}} = \left(6.67\times10^{-12}\right)\cos 2\pi\left(\frac{x}{3.33} - 90.0\times10^{6}t\right)\hat{\mathbf{k}}}$

where $\vec{\mathbf{E}}$ is in V/m, $\vec{\mathbf{B}}$ in tesla, x in meters, and t in seconds.

(e) $$I = \frac{E_{\max}^2}{2\mu_0 c} = \frac{\left(2.00\times10^{-3}\ \text{V/m}\right)^2}{2\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}$$

$$= \boxed{5.31\times10^{-9}\ \text{W/m}^2}$$

(f) From Equation 34.26, $I = cu_{\text{avg}}$ so $u_{\text{avg}} = \frac{I}{c} = \boxed{1.77\times10^{-17}\ \text{J/m}^3}$

(g) From Equation 34.30, the pressure is

$$P = \frac{2I}{c} = \frac{(2)\left(5.31\times10^{-9}\ \text{W/m}^2\right)}{3.00\times10^{8}\ \text{m/s}} = \boxed{3.54\times10^{-17}\ \text{Pa}}$$

P34.77 (a) The magnetic field has amplitude

$$B_{\max} = \frac{E_{\max}}{c} = \frac{175\ \text{V/m}}{3.00\times10^{8}\ \text{m/s}} = 5.83\times10^{-7}\ \text{T} = \boxed{583\ \text{nT}}$$

(b) The wave number is

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.015\ 0\ \text{m}} = \boxed{419\ \text{m}^{-1}}$$

(c) The angular frequency is

$$\omega = kc = \left(419\ \text{m}^{-1}\right)\left(3.00\times10^{8}\ \text{m/s}\right) = 1.26\times10^{11}\ \text{s}^{-1}$$

(d) $\vec{\mathbf{S}} \propto \vec{\mathbf{E}}\times\vec{\mathbf{B}}$, $\vec{\mathbf{S}}$ is in the x direction, and $\vec{\mathbf{E}}$ vibrates in the y direction (xy plane), so $\vec{\mathbf{B}}$ must vibrate in the z direction, thus $\boxed{\vec{\mathbf{B}} \text{ vibrates in the } xz \text{ plane}}$.

(e) The magnitude of the average Poynting vector is the wave intensity

$$S_{\text{avg}} = \frac{E_{\max}B_{\max}}{2\mu_0} = \frac{\left(175\ \text{V/m}\right)\left(5.83\times10^{-7}\ \text{T}\right)}{2\left(4\pi\times10^{-7}\ \text{N/A}^2\right)} = 40.6\ \text{W/m}^2$$

The Poynting vector itself points in the direction of energy transport:

$$\vec{\mathbf{S}}_{\text{avg}} = \boxed{40.6\hat{\mathbf{i}}\ \text{W/m}^2}$$

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(f) For perfect reflection, the pressure is

$$P_r = \frac{2S}{c} = \frac{2\left(40.6\ \text{W/m}^2\right)}{3.00\times10^8\ \text{m/s}} = 2.71\times10^{-7}\ \text{N/m}^2 = \boxed{271\ \text{nPa}}$$

(g) From Newton's second law,

$$a = \frac{\sum F}{m} = \frac{PA}{m} = \frac{\left(2.71\times10^{-7}\ \text{N/m}^2\right)\left(0.750\ \text{m}^2\right)}{0.500\ \text{kg}}$$

$$= 4.07\times10^{-7}\ \text{m/s}^2$$

$$\vec{\mathbf{a}} = \boxed{407\hat{\mathbf{i}}\ \text{nm/s}^2}$$

P34.78 We can approximate the magnetic field as uniform over the area of the loop while it oscillates in time as $B = B_{\max}\cos\omega t$. The induced voltage is

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}(BA\cos\theta) = -A\frac{d}{dt}\left(B_{\max}\cos\omega t\cos\theta\right)$$

or $$\mathcal{E} = AB_{\max}\,\omega(\sin\omega t\cos\theta)$$

(a) Since the angular frequency is $\omega = 2\pi f$, and the area of the loop is πr^2, the amplitude of this emf is

$$\boxed{\mathcal{E}_{\max} = 2\pi^2 r^2 f B_{\max}\cos\theta}$$

where θ is the angle between the magnetic field and the normal to the loop.

(b) If $\vec{\mathbf{E}}$ is vertical, $\vec{\mathbf{B}}$ is horizontal, so the plane of the loop should be vertical and the plane should contain the line of sight of the transmitter.

P34.79 (a) From the particle under a net force model, the acceleration of the astronaut is

$$a = \frac{F}{m} = \frac{1}{m}\frac{dp}{dt}$$

where dp/dt is the rate of change of momentum of the astronaut. From the momentum version of the isolated system model, the rate of change of momentum of the astronaut is equal in magnitude to that of the radiation from the flashlight. The momentum of the radiation leaving the flashlight can be evaluated from Equation 34.27, assuming the same equation for complete absorption applies to complete emission. Therefore, the acceleration of the astronaut can be written as

$$a = \frac{1}{m}\frac{d}{dt}\left(\frac{T_{\text{ER}}}{c}\right) = \frac{1}{mc}\frac{dT_{\text{ER}}}{dt} = \frac{P}{mc}$$

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where P is the power of the radiation leaving the flashlight. Because all three variables on the right side of this equation are constant, the acceleration of the astronaut is constant and we can use the particle under constant acceleration model. The position of the astronaut is given by,

$$x = x_i + v_i t + \frac{1}{2}at^2 = 0 + 0 + \frac{1}{2}\left(\frac{P}{mc}\right)t^2$$

where we have defined the initial position of the astronaut as $x = 0$ and recognized that the astronaut begins from rest. Solve for the time at which the astronaut is at a position x:

$$t = \sqrt{\frac{2mcx}{P}}$$

Substituting numerical values,

$$t = \sqrt{\frac{2(110\text{ kg})(3.00\times10^8\text{ m/s})(10.0\text{ m})}{100\text{ W}}} = 8.12\times10^4\text{ s}$$

$$= \boxed{22.6\text{ h}}$$

(b) There are no external forces on the astronaut–flashlight system, so the system is isolated for momentum. Apply the conservation of momentum principle along an axis parallel to the direction of travel of the astronaut and the flashlight:

$$\Delta p = 0 \quad \rightarrow \quad p_i = p_f \quad \rightarrow \quad 0 = \left(m - m_f\right)v - m_f\left(v_{\text{rel}} - v\right)$$

Solve for the speed of the astronaut:

$$v = \left(\frac{m_f}{m}\right)v_{\text{rel}}$$

Because this speed is constant, we can use the particle under constant velocity model to find the time interval required for the astronaut to arrive back at her spacecraft:

$$\Delta t = \frac{\Delta x}{v} = \frac{m}{m_f}\left(\frac{\Delta x}{v_{\text{rel}}}\right)$$

Substituting numerical values,

$$\Delta t = \left(\frac{110\text{ kg}}{3.00\text{ kg}}\right)\frac{10.0\text{ m}}{12.0\text{ m/s}} = \boxed{30.6\text{ s}}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P34.2 (a) $7.19 \times 10^{11}\ \text{V/m} \cdot \text{s}$; (b) 2.00×10^{-7} T

P34.4 $\left(-4.39\hat{\mathbf{i}} - 1.76\hat{\mathbf{j}}\right) \times 10^{11}\ \text{m/s}^2$

P34.6 (a) $3.15 \times 10^{3}\hat{\mathbf{j}}$ N/C; (b) $5.25\hat{\mathbf{k}} \times 10^{-7}$ T; (c) $4.83\left(-\hat{\mathbf{j}}\right) \times 10^{-16}$ N

P34.8 11.0 m

P34.10 4.738×10^{14} Hz

P34.12 733 nT

P34.44 60.0 km

P34.16 See P34.16 for full explanation.

P34.18 The ratio of ω to k is higher than the speed of light in a vacuum, so the wave as described is impossible.

P34.20 8.64×10^{10} m

P34.22 (a) 6.75 W/m^2; (b) 6.64×10^{3} W/m^2; (c) A powerful automobile running on sunlight would have to carry on its roof a solar panel that is huge compared to the size of the car; (d) Agriculture and forestry for food and fuels, space heating of large and small buildings, water heating, and heating for drying and many other processes are current and potential applications of solar energy.

P34.24 (a) 0; (b) $\left(11.5\hat{\mathbf{i}} - 28.6\hat{\mathbf{j}}\right)$ W/m^2

P34.26 For the small container, 33.4° and for the larger container, 21.7°

P34.28 (a) 88.8 nW/m^2; (b) 11.3 MW

P34.30 (a) 5.16×10^{-10} T; (b) Since the magnetic field of the Earth is approximately 5×10^{-10} T, the Earth's field is some 100 000 times stronger.

P34.32 5.16 m

P34.34 (a) 540 V/m; (b) 2.58 μJ/m^3; (c) 773 W/m^2

P34. 36 83.3 nPa

P34.38 (a) $\sqrt{\dfrac{2\mu_0 cP}{\pi r^2}}$; (b) $\dfrac{P\ell}{c}$; (c) $\dfrac{P\ell}{c^2}$

P34.40 (a) 5.82×10^{8} N; (b) 6.10×10^{13} times stronger

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P34.42 (a) 590 W/m^2; (b) 2.10×10^{16} W; (c) 7.01×10^7 N; (d) ~10^{13} times stronger; (e) The values are similar for both planets because both the forces follow inverse-square laws. The force ratios are not identical for the two planets because of their different radii and masses.

P34.44 (a) 1.00×10^3 km or 621 mi; (b) While the project may be theoretically possible, it is not very practical.

P34.46 (a) $\frac{1}{2}\mu_0 c J_{\max} \cos(kx - \omega t)\hat{\mathbf{j}}$; (b) $\frac{1}{4}\mu_0 c J_{\max}^2 \cos^2(kx - \omega t)\hat{\mathbf{i}}$; (c) $\frac{1}{8}\mu_0 c J_{\max}^2$; (d) 3.48 A/m

P34.48 $\dfrac{2\pi mc}{qB}$

P34.50 (a) ~10^8 Hz radio wave; (b) ~10^{13} Hz infrared

P34.52 Listeners 100 km away will receive the news before the people in the newsroom.

P34.54 See table in P34.54 for full description.

P34.56 95.1 mV/m

P34.58 $E = 300\cos(62.8x - 1.88 \times 10^{10} t)$ and $B = 1.00\cos(62.8x - 1.88 \times 10^{10} t)$

P34.60 (a) 1.50 cm; (b) 25.0 μJ; (c) 7.37 mJ/m^3; (d) $E_{\max}$ = 40.8 kV/m, $B_{\max}$ = 136 μT; (e) 83.3 μN

P34.62 (a) 93.3%; (b) 50.0%; (c) 0

P34.64 $\dfrac{3SR^2}{4cGM_S\rho}$

P34.66 (a) 6.30×10^{-6} Pa; (b) 1.60×10^{10} times smaller than atmospheric pressure

P34.68 (a) $\vec{\mathbf{E}} = \dfrac{38.8}{r^2}\hat{\mathbf{r}}$, where $\vec{\mathbf{E}}$ is in volts per meter and r is in meters;

(b) $E_{\max} = \dfrac{38.7}{r}$ where E is in volts per meter and r is in meters;

(c) 12.9 μm, but the expression in part (b) does not apply if this point is inside the source; (d) From part (c), we see that in the radiated wave, the field amplitude is inversely proportional to distance. As the distance doubles, the amplitude is cut in half; (e) In the static case, the field is inversely proportional to the square of distance. As the distance doubles, the field is reduced by a factor of 4.

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P34.70 (a) See ANS. FIG. P34.70; (b) $\frac{1}{2}\epsilon_0 E_{\max}^2 \cos^2(kx)$; (c) $\frac{1}{2\mu_0} B_{\max}^2 \cos^2(kx)$;
(d) $\epsilon_0 E_{\max}^2 \cos^2(kx)$; (e) $\frac{1}{2}\epsilon_0 E_{\max}^2 \lambda A$; (f) $\frac{1}{2}\epsilon_0 c E_{\max}^2 A$; (g) $\frac{1}{2}\epsilon_0 c E_{\max}^2$;
(h) The result in part (g) agrees with $I = \frac{E_{\max}^2}{2\mu_0 c}$ in Equation 34.24.

P34.72 (a) $\frac{4}{3}\rho c g r$; (b) $\frac{4}{3}\pi\rho c g r^3$

P34.74 (a) See P34.74(a) for full proof; (b) 1.76×10^{13} m/s^2; (c) 1.75×10^{-27} W;
(d) 1.80×10^{-24} W

P34.76 (a) 3.33 m; (b) 11.1 ns; (c) 6.67 pT;
(d) $\vec{\mathbf{E}} = \left(2.00 \times 10^{-3}\right)\cos 2\pi \left(\frac{x}{3.33} - 90.0 \times 10^6 t\right)\hat{\mathbf{j}}$ and
$\vec{\mathbf{B}} = \left(6.67 \times 10^{-12}\right)\cos 2\pi\left(\frac{x}{3.33} - 90.0 \times 10^6 t\right)\hat{\mathbf{k}}$; (e) 5.31×10^{-9} W/m^2;
(f) 1.77×10^{-17} J/m^2; (g) 3.54×10^{-17} Pa

P34.78 (a) $\mathcal{E}_{\max} = 2\pi^2 r^2 f B_{\max} \cos\theta$; (b) The plane of the loop should be vertical and the plane should contain the line of sight of the transmitter.

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